

APPENDIX U

Secondary Battery And Method For Manufacturing The Same
Transaction History

Date	Transaction Description
05-14-2025	Information Disclosure Statement (IDS) Filed
05-14-2025	Applicants have given acceptable permission for participating foreign
05-14-2025	PTO-SB-69-Authorize EPO Access to Search Results
05-14-2025	Patent Term Adjustment - Ready for Examination
05-14-2025	Information Disclosure Statement (IDS) Filed
05-14-2025	Entity Status Set To Undiscounted (Initial Default Setting or Status Change)
05-14-2025	Initial Exam Team nn
05-22-2025	Application ready for PDX access by participating foreign offices
05-22-2025	Case Docketed to Examiner in GAU
05-22-2025	Application Dispatched from OIPE
05-22-2025	Sent to Classification Contractor
05-22-2025	FITF set to YES - revise initial setting
05-23-2025	Email Notification
05-23-2025	Application Is Now Complete
05-23-2025	Mail Pre-Exam Notice
05-23-2025	Filing Receipt
08-29-2025	Email Notification
08-29-2025	PG-Pub Issue Notification
01-16-2026	Information Disclosure Statement (IDS) Filed
01-16-2026	Information Disclosure Statement (IDS) Filed



UNITED STATES
PATENT AND TRADEMARK OFFICE

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/207,758	05/14/2025 01:37:38 PM Z ET	LGES 3.3F-1654 CCCC II

Title of Invention

Secondary Battery And Method For Manufacturing The Same

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	2553	FILED BY	Sharon McGee
PATENT CENTER #	70427322	FILING DATE	-
CUSTOMER #	86765	FIRST NAMED INVENTOR	Jae Woong Kim
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Kevin Kocun

Documents

TOTAL DOCUMENTS: 10

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
LGES6209 - Information Disclosure Statement Fillable PDF.pdf	11	Information Disclosure Statement (IDS) Form (SB08)	1317 KB
LGES6209 - IDS - Information Disclosure Statement - Reference in Previous Application LD-4.pdf	1	Transmittal Letter	21 KB
LGES6209 - Communication Regarding Certified Copy.pdf	1	Miscellaneous Incoming Letter	21 KB
LGES6209 - Power of Attorney by Applicant PTO AIA-82.pdf	1	Power of Attorney	17 KB
LGES6209 - Application Data	9	Application Data Sheet	2174 KB

Sheet Fillable PDF.pdf			
LGES6209 - Transmittal -- Utility Patent Application PTO AIA-15.pdf	1	Transmittal of New Application	33 KB
POA - LG Energy Solution Ltd.pdf	1	Power of Attorney	46 KB
Declaration.pdf	2	Oath or Declaration filed	100 KB
Specification LGES 33F-1654 CCCC II-APP.TEXT.docx	20	Application body structured text document	37 KB
Warning: Document has [Balance SBCS characters and DBCS characters] option enabled and may result in reduction of pages, however, no data or text has been modified.			
Drawings.pdf	9	Drawings-only black and white line drawings	765 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
LGES6209 - Information Disclosure Statement Fillable PDF.pdf	EA71CC7660BB4A92899D20E0B6DF7DBB178AA6D919B035413CE0E0495F091F20BBAD2FAC4EB4A141A51B455E264B21D295EEBFFBEE023434D07ADCF9BC181691
LGES6209 - IDS - Information Disclosure Statement - Reference in Previous Application LD-4.pdf	34DA43F1C3742E8404668424EB78A34E66D88A4833A73BBBAF05BBD6DDF2FF4A772FAA22C6CE61AEFAA53CFC44A919F1D5975733B5A962ED5BC3789325417132
LGES6209 - Communication Regarding Certified Copy.pdf	78B605F5C27D20D84FC9C6184C7632DD26EAEDB98392DBDFBCCC37266A34AD1F6D17F7BDD4C91169A616806AC46B18830D86C1D714D6378603BE59D7FD53C23E
LGES6209 - Power of Attorney by Applicant PTO AIA-82.pdf	416AC2D7C4861E79B1D69825EC2AD5FD1807B76834A4EA598EFBF839E0F0C0662117F26A504DF9C9091DA26AD81A7BBA2B10F1E7C3F632CCC5B44C8268706400
LGES6209 - Application Data Sheet Fillable PDF.pdf	F5D535CEF9D4CB3AA91F8BC474E8BBCB3F1DA96EEAA0DEC642BC8106500A7E2FC895F4A20D4CEF0D75D5519B1B8496C7CEE0BCB452756BF3B2248D1E2DCE241E
LGES6209 - Transmittal -- Utility Patent Application PTO AIA-15.pdf	68BEFB67F6AA0817203F84707D4293B80C514D28C4404D38D61B212E5E37929B0F8328ADDC4B0438BDAB2F6D019C9035DFE635D67A87D93C5A9AF9A634783515
POA - LG Energy Solution	F18C66A48E1D0E420627962F7063C6E831339EDFBCB6123EE

Ltd.pdf	3E1ACD09BE0C007FD748A528FF87D98D94170D7E971E26A7E0AEF38D0F53940B2E4D37C2805D84F
Declaration.pdf	15E8E267EEA2282A6243812DB47489BEE5B8983900EC30C5522760E1A5FB020DD18B5BD702ECB2B185825816DD6D9A6C16A2E655660DFC39856A0F8F6E84E66C
Specification LGES 33F-1654 CCCC II-APP.TEXT.docx	B4ED4B34DAB72E2BF9173E68185E92442D7B360A608FAF803B895C909D785FC357D6EC0AA09210D951EC0D5B99EF2164016D3927D31F4ABEF91FF19917DD0466
Drawings.pdf	F7D9599201823ECDB36B7F3197457C4A7CB48F780C5149EF5A080A88682E8058BB971B10ACFB178C4CEBA290422BACAA9A674A9E97E34C252606404D07AC119C

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT

Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit		N/A
Examiner Name	Not Yet Assigned	
Attorney Docket Number		LGES 3.3F-1654 CCCC II

TIMING STATEMENT

Please see 37 CFR 1.97(e) to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

- ☐ See attached timing statement under 37 CFR 1.97(e).
- ☒ A timing statement under 37 CFR 1.97(e) is not submitted herewith.

TIMING FEE

☐ The fee set forth in 37 CFR 1.17(p) has been submitted herewith.

COPIES

☒ An identification of an earlier application pursuant to 37 CFR 1.98(d)(1) is attached.

SIZE FEE ASSERTION

Please see 37 CFR 1.17(v) and the [IDS Auto-Load Instructions](#) for completing this form to make the appropriate selection of an assertion under 37 CFR 1.98. For the information disclosure statement (IDS) submitted herewith, the applicant or patent owner certifies the following with respect to the cumulative number of applicant-provided or patent-owner provided items of information submitted to date including those in the accompanying IDS (select only one):

- ☐ No IDS size fee is required under 37 CFR 1.17(v) at this time.
- ☒ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(1).
- ☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(2).
- ☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(3).

SIGNATURE

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 11.18. Please see 37 CFR 1.4(d) for the form of the signature.

Signature	/Joseph H. Karlin/	Date (YYYY-MM-DD)	2025-05-14
Name/Print	Joseph H. Karlin	Registration Number	62,471

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT

Application Number		
Filing Date		
First Named Inventor	Jae Woong Kim	
Art Unit		N/A
Examiner Name	Not Yet Assigned	
Attorney Docket Number		LGES 3.3F-1654 CCCC II

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		
	Filing Date		
	First Named Inventor	Jae Woong Kim	
	Art Unit		N/A
	Examiner Name	Not Yet Assigned	
	Attorney Docket Number		LGES 3.3F-1654 CCCC II

U.S. PATENTS							Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Inventor of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	
	1	3761314		1973-09-25	CAILLEY J		
	2	4822377		1989-04-18	Wolff		
	3	6569565		2003-05-27	Ligeois et al.		
	4	11652232		2023-05-16	Kim et al.		
	5	11973177		2024-04-30	Kim et al.		
If you wish to add additional U.S. Patent citation information, please click the Add button.							Add
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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Inventor or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	
	1	20010023038	A1	2001-09-20	Ligeois et al.		

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STATEMENT BY APPLICANT

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Art Unit		N/A
Examiner Name	Not Yet Assigned	
Attorney Docket Number		LGES 3.3F-1654 CCCC II

	2	20050003264	A1	2005-01-06	Oh et al.	
	3	20050142436	A1	2005-06-30	Arai et al.	
	4	20050277022	A1	2005-12-15	Kozuki, Kiyomi	
	5	20050287428	A1	2005-12-29	Cheon et al.	
	6	20050287428	A1	2005-12-29	Cheon et al.	
	7	20050287432	A1	2005-12-29	Cheon et al.	
	8	20060024572	A1	2006-02-02	Lee, Sang-Won	
	9	20060127774	A1	2006-06-15	Kim et al.	
	10	20070196730	A1	2007-08-23	Kozuki, Kiyomi	
	11	20100081052	A1	2010-04-01	MORISHIMA et al.	
	12	20100190056	A	2010-07-29	Turner	

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Attorney Docket Number		LGES 3.3F-1654 CCCC II

	13	20100310926	A1	2010-12-09	WANG et al.	
	14	20110067227	A1	2011-03-24	Sohn	
	15	20110293977	A1	2011-12-01	Kim et al.	
	16	20120009450	A1	2012-01-12	Chun	
	17	20120208055	A1	2012-08-16	Ahn	
	18	20130108903	A1	2013-05-02	Martin et al.	
	19	20130252054	A1	2013-09-26	BARONE et al.	
	20	20130344364	A1	2013-12-26	Nanaumi et al.	
	21	20140050956	A1	2014-02-20	HUANG	
	22	20150295270	A1	2015-10-15	Chun, Kwan-Sic	
	23	20160351966	A1	2016-12-01	Batson et al.	

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	Examiner Name	Not Yet Assigned		
	Attorney Docket Number		LGES 3.3F-1654 CCCC II	

	24	20180145304	A1	2018-05-24	KOGETSU et al.	
	25	20190379028	A1	2019-12-12	Lim et al.	
	26	20240243335	A1	2024-07-18	Kim et al.	

If you wish to add additional U.S. Published Application citation information, please click the Add button

FOREIGN PATENT DOCUMENTS Remove								
Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1	101714624	CN	A	2010-05-26	Guangzhou Fullriver Battery New Technology Co., Lt	Machine generated translation is attached.	☒
	2	102024936	CN	A	2011-04-20	Samsung Sdi Co., Ltd	U.S. Publication No. 20110067227 is an English language counterpart which was previously cited.	☒
	3	201450072	CN	U	2010-05-05	Dongguan AmpereX Electronics Technology Co., Ltd	U.S. Publication No. 20100310926 is an English language counterpart which was previously cited.	☒
	4	203250828	CN	U	2013-10-23	A123 Systems, Inc	U.S. Publication No. 20130108903 is an English language counterpart which was previously cited.	☒
	5	204577523	CN	U	2015-08-19	Contemporary AmpereX Technology Ltd	Machine generated translation is attached.	☒
	6	1134820	EP	A1	2001-09-19	Cit Alcatel	U.S. Publication No. 20010023038 A1 and U.S. Patent No. 6,569,565 are English language counterparts.	☒

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First Named Inventor	Jae Woong Kim	
Art Unit		N/A
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Attorney Docket Number		LGES 3.3F-1654 CCCC II

	7	2000-077078	JP	A	2000-03-14	Matsushita Electric Ind Co Ltd	Machine generated translation is attached.	☒
	8	2000-228182	JP	A	2000-08-15	Toyota Central Res & Dev	Machine generated translation is attached.	☒
	9	2003338276	JP	A	2003-11-28	Matsushita Electric Ind Co Ltd	Machine Generated Translation attached.	☒
	10	2005276459	JP	A	2005-10-06	Sanyo Electric Co	English translation of Abstract only.	☒
	11	2006128106	JP	A	2006-05-18	Samsung Sdi Co Ltd	U.S. Publication No. 20060127774 is an English language counterpart.	☒
	12	2006-324049	JP	A	2006-11-30	Sony Corp	Machine generated translation is attached.	☒
	13	2007227137	JP	A	2007-09-06	Matsushita Electric Ind Co Ltd	U.S Publication No. 2007196730 is an English language counterpart.	☒
	14	2008066040	JP	A	2008-03-21	Sony Corp	Machine generated translation is attached.	☒
	15	2011065981	JP	A	2011-03-31	Samsung Sdi Co Ltd	U.S Publication No. 2011067227 is an English language counterpart.	☒
	16	2016001575	JP	A	2016-01-07	Denso Corp	Machine generated translation is attached.	☒
	17	2020509534	JP	A	2020-03-26	LG CHEM Ltd.	U.S Publication No. 2019379028 is an English language counterpart.	☒

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Art Unit		N/A
Examiner Name	Not Yet Assigned	
Attorney Docket Number		LGES 3.3F-1654 CCCC II

	18	4401634	JP	B2	2010-01-20	Matsushita Electric Ind Co Ltd	Machine generated translation is attached.	☒
	19	5117133	JP	B2	2013-01-09	Hitachi Vehicle Energy Ltd	Machine generated translation is attached.	☒
	20	5790576	JP	B2	2015-10-07	Toyota Industries Corp	Machine generated translation is attached.	☒
	21	6045286	JP	B2	2016-12-14	Hitachi Vehicle Energy Ltd	Machine generated translation is attached.	☒
	22	H0935701	JP	A	1997-02-07	Denso Corp	Machine generated translation is attached.	☒
	23	H11339756	JP	A	1999-12-10	Toyota Central Res & Dev	Machine generated translation is attached.	☒
	24	100515833	KR	B1	2005-09-21	Samsung Sdi Co., Ltd	U.S. Publication No. 20050003264 A1 is an English language counterpart.	☒
	25	100599710	KR	B1	2006-07-12	Samsung Sdi Co., Ltd	U.S. Publication No. 20060024572 A1 is an English language counterpart.	☒
	26	100599749	KR	B1	2006-07-12	Samsung Sdi Co., Ltd	U.S. Publication No. 20050287428 A1 is an English language counterpart.	☒
	27	20000051740	KR	A	2000-08-16	Samsung Sdi Co., Ltd	Machine generated translation is attached.	☒
	28	20050121904	KR	A	2005-12-28	Samsung Sdi Co., Ltd	U.S. Publication No. 20050287428 A1 is an English language counterpart.	☒

INFORMATION DISCLOSURE
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Art Unit		N/A
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Attorney Docket Number		LGES 3.3F-1654 CCCC II

	29	20050121914	KR	A	2005-12-28	Samsung Sdi Co., Ltd	U.S. Publication No. 20050287432 A1 is an English language counterpart.	☒
	30	20060022358	KR	A	2006-03-10	Samsung Sdi Co., Ltd	Machine generated translation is attached.	☒
	31	20100071941	KR	A	2010-06-29	Lg Chemical Ltd	U.S. Publication No. 20110293977 A1 is an English language counterpart.	☒
	32	20120006389	KR	A	2012-01-18	Samsung Sdi Co., Ltd	U.S. Publication No. 20120009450 A1 is an English language counterpart.	☒
	33	20120092367	KR	A	2012-08-21	Samsung Sdi Co., Ltd	U.S. Publication No. 20120208055 A1 is an English language counterpart.	☒
	34	20130073329	KR	A	2013-07-03	Datechnology Co., Ltd	Machine generated translation is attached.	☒
	35	2015-0117135	Kr	A	2015-10-19	Samsung Sdi Co Ltd	U.S. Publication No. 20150295270 A1 is an English language counterpart.	☒
	36	2016-0009406	KR	A	2016-01-26	Lg Chemical Ltd	Machine generated translation is attached.	☒
	37	20160113612	KR	A	2016-09-30	A123 Systems LLC	U.S. Publication No. 20160351966 A1 is an English language counterpart.	☒
	38	20170006437	KR	A	2017-01-18	Lg Chemical Ltd	Machine generated translation is attached.	☒
	39	2012086690	WO	A1	2012-06-28	Hitachi Vehicle Energy, Ltd	U.S Publication No. 2013344364 is an English language counterpart.	☒

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number			
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	Attorney Docket Number	LGES 3.3F-1654 CCCC II		

	40	2014188501	WO	A1	2014-11-27	Hitachi Automotive Systems, Ltd	Machine generated translation is attached.	☒
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NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	International Search Report from Application No. PCT/KR2018/004366 mailed July 20, 2018, pages 1 - 2	☒
	2	Extended European Seach Report including Written Opinion for Application No. EP18784869.2 dated January 21, 2020, 5 pages.	☐
	3	Chiinese Search Report for Application No. 201880025247, dated August 24, 2021, 2 pages.	☐
	4	Extended European Seach Report including Written Opinion for Application No. 24187569.9 dated 10/11/24. 9 pages.	☐
	5	Extended European Seach Report including Written Opinion for Application No. 24187669.7 dated 10/18/24. 9 pages.	☐

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EXAMINER SIGNATURE

Examiner Signature		Date Considered	
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*EXAMINER: Initial if reference considered and whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

Other Prior Art

According to the information contained in form PTO-1449 or PTO-892, there are one or more other prior art/non-patent literature documents missing from the original file history record obtained from the United States Patent and Trademark Office. Upon your request we will attempt to obtain these documents from alternative resources. Please note that additional charges will apply for this service.

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: May 14, 2025
Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Docket No. LGES 3.3F-1654 CCCC II
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: Not Yet Assigned

Confirmation No.: N/A

Filed: Concurrently Herewith

Art Unit: N/A

For: Secondary Battery And Method For
Manufacturing The Same

Examiner: Not Yet Assigned

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Dear Commissioner:

It is respectfully requested that the references listed on the enclosed form be made of record and considered with respect to the above-referenced U.S. patent application. A copy of each foreign patent document and each non-patent literature document was of record in Application Nos. 18/620,269, 18/618,609; 18/132,719; and 16/498,751, the benefit of which is claimed under § 120. Submission of the present Information Disclosure Statement should not be taken as an admission that the cited references are legally available prior art or that the same are pertinent or material or even analogous art to the claimed invention.

The required fees are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments to our Deposit Account No. 12-1095.

Dated: May 14, 2025

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/
Joseph H. Karlin
Registration No.: 62,471
LERNER DAVID LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: May 14, 2025

Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Docket No. LGES 3.3F-1654 CCCCII
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: Not Yet Assigned

Confirmation No.: N/A

Filed: Concurrently Herewith

Art Unit: N/A

For: Secondary Battery And Method For
Manufacturing The Same

Examiner: Not Yet Assigned

COMMUNICATION REGARDING CERTIFIED COPY

Dear Commissioner:

The present application claims priority to one or more prior foreign applications under 35 U.S.C. § 119(b). Applicant hereby identifies prior-filed nonprovisional Application No. 16/498,751, for which a benefit is claimed under 35 U.S.C. §§ 120, 121, 365(c), or 386(c), as containing a certified copy of such foreign application(s). Accordingly, pursuant to 37 C.F.R. § 1.55(h), the requirement for timely filing a certified copy of the foreign application(s) is considered satisfied in this application.

All required fees are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments to our Deposit Account No. 12-1095.

Dated: May 14, 2025

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/
Joseph H. Karlin

Registration No.: 62,471
LERNER DAVID LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS			
NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.			
Application Number	Not Yet Assigned		
Filing Date	Concurrently Herewith		
First Named Inventor	Jae Woong Kim		
Title	Secondary Battery And Method For Manufacturing The Same		
Art Unit	N/A		
Examiner Name	Not Yet Assigned		
Attorney Docket Number	LGES 3.3F-1654 CCCCII		
SIGNATURE of Applicant or Patent Practitioner			
Signature	/Joseph H. Karlin/	Date (Optional)	May 14, 2025
Name	Joseph H. Karlin	Registration Number	62,471
Title (if Applicant is a juristic entity)			
Applicant Name (if Applicant is a juristic entity)			
NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.			
☐	*Total of 1 forms are submitted.		

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).	
Dated: May 14, 2025	Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES 3.3F-1654 CCCC II
		Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Inventor Information:

Inventor 1

Remove

Legal Name

Prefix	Given Name	Middle Name	Family Name	Suffix
	Jae Woong		Kim	

Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service

City	Daejeon	Country of Residence ⁱ	KR
------	---------	-----------------------------------	----

Mailing Address of Inventor:

Address 1	LG Energy Solution Research Park			
Address 2	188, Munji-ro, Yuseong-gu			
City	Daejeon	State/Province		
Postal Code	34122	Country ⁱ	KR	

Inventor 2

Remove

Legal Name

Prefix	Given Name	Middle Name	Family Name	Suffix
	Hyoung Kwon		Kim	

Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service

City	Daejeon	Country of Residence ⁱ	KR
------	---------	-----------------------------------	----

Mailing Address of Inventor:

Address 1	LG Energy Solution Research Park			
Address 2	188, Munji-ro, Yuseong-gu			
City	Daejeon	State/Province		
Postal Code	34122	Country ⁱ	KR	

Inventor 3

Remove

Legal Name

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES 3.3F-1654 CCCC II
		Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same		

Prefix	Given Name	Middle Name	Family Name	Suffix
	Sang Uck		Kim	
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service				
City	Daejeon	Country of Residence ⁱ	KR	

Mailing Address of Inventor:

Address 1	LG Energy Solution Research Park			
Address 2	188, Munji-ro, Yuseong-gu			
City	Daejeon	State/Province		
Postal Code	34122	Country ⁱ	KR	
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.				

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below.
For further information see 37 CFR 1.33(a).

☐ An Address is being provided for the correspondence Information of this application.

Customer Number	86765		
Email Address		Add Email	Remove Email

Application Information:

Title of the Invention	Secondary Battery And Method For Manufacturing The Same		
Attorney Docket Number	LGES 3.3F-1654 CCCC II	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	9	Suggested Figure for Publication (if any)	1

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country ⁱ

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES 3.3F-1654 CCCC II
		Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same		

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	86765		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove			
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
	Continuation of	18/620269	2024-03-28		
Prior Application Status	Pending	Remove			
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
18/620269	Continuation of	18/618609	2024-03-27		
Prior Application Status	Patented	Remove			
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
18/618609	Continuation of	18/132719	2023-04-10	11973177	2024-04-30
Prior Application Status	Patented	Remove			
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
18/132719	Continuation of	16/498751	2019-09-27	11652232	2023-05-16

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES 3.3F-1654 CCCC II
		Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same		

Prior Application Status		Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
16/498751	a 371 of international	PCT/KR2018/004366	2018-04-13
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
10-2018-0043191	KR	2018-04-13	

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
10-2017-0048645	KR	2017-04-14	
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐

This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	LGES 3.3F-1654 CCCC II
	Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same	

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People’s Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO’s Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

- ☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.
- ☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	LGES 3.3F-1654 CCCC II
	Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same	

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant 1

Remove

If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.

Clear

☒ Assignee

☐ Legal Representative under 35 U.S.C. 117

☐ Joint Inventor

☐ Person to whom the inventor is obligated to assign.

☐ Person who shows sufficient proprietary interest

If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:

Name of the Deceased or Legally Incapacitated Inventor:

If the Applicant is an Organization check here. ☒

Organization Name

LG Energy Solution, Ltd.

Mailing Address Information For Applicant:

Address 1

Tower 1, 108, Yeoui-daero, Yeongdeungpo-gu

Address 2

City

Seoul

State/Province

Country

KR

Postal Code

07335

Phone Number

Fax Number

Email Address

Additional Applicant Data may be generated within this form by selecting the Add button.

Add

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES 3.3F-1654 CCCC II
		Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same		

Assignee 1

Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.

Remove

If the Assignee or Non-Applicant Assignee is an Organization check here.

X

Organization Name

LG Energy Solution, Ltd.

Mailing Address Information For Assignee including Non-Applicant Assignee:

Address 1

Tower 1, 108, Yeoui-daero, Yeongdeungpo-gu

Address 2

City

Seoul

State/Province

Countryⁱ

KR

Postal Code

07335

Phone Number

Fax Number

Email Address

Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.

Add

Signature:

Remove

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the “Authorization or Opt-Out of Authorization to Permit Access” section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet must be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, all joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of all joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature

/Joseph H. Karlin/

Date (YYYY-MM-DD)

2025-05-14

First Name

Joseph

Last Name

Karlin

Registration Number

62,471

Additional Signature may be generated within this form by selecting the Add button.

Add

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	LGES 3.3F-1654 CCCC II
		Application Number	
Title of Invention	Secondary Battery And Method For Manufacturing The Same		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

PTO/AIA/15 (10-17)
Approved for use through 11/30/2027. OMB 0651-0032
U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995 no persons are required to respond to a collection of information unless it displays a valid OMB control number.

UTILITY PATENT APPLICATION TRANSMITTAL (Only for new nonprovisional applications under 37 CFR 1.53(b))	Attorney Docket No. LGES 3.3F-1654 CCCC II	
	First Named Inventor Jae Woong Kim	
	Title	Secondary Battery And Method For Manufacturing The Same
	Priority Mail Express® Label No.	

APPLICATION ELEMENTS See MPEP chapter 600 concerning utility patent application contents. 1. ☐ Fee Transmittal Form (PTO/SB/17 or equivalent) 2. ☐ Applicant asserts small entity status. See 37 CFR 1.27 3. ☐ Applicant certifies micro entity status. See 37 CFR 1.29. Applicant must attach form PTO/SB/15A or B or equivalent. 4. ☒ Specification [Total Pages 20] Both the claims and abstract must start on a new page. (See MPEP § 608.01(a) for information on the preferred arrangement) 5. ☒ Drawing(s) (35 U.S.C. 113) [Total Sheets 9] 6. Inventor's Oath or Declaration [Total Pages 2] (including substitute statements under 37 CFR 1.64 and assignments serving as an oath or declaration under 37 CFR 1.63(e)) a. ☐ Newly executed (original or copy) b. ☒ A copy from a prior application (37 CFR 1.63(d)) 7. ☒ Application Data Sheet * See note below. See 37 CFR 1.76 (PTO/AIA/14 or equivalent) 8. ☐ CD-ROM or CD-R in duplicate, large table, or Computer Program (Appendix) ☐ Landscape Table on CD 9. Nucleotide and/or Amino Acid Sequence Submission (if applicable, items a. – c. are required) a. ☐ Computer Readable Form (CRF) b. ☐ Specification Sequence Listing on: i. ☐ CD-ROM or CD-R (2 copies); or ii. ☐ Paper c. ☐ Statements verifying identity of above copies ADDRESS TO: Commissioner for Patents P.O. Box 1450 Alexandria, VA 22313-1450 ACCOMPANYING APPLICATION PAPERS 10. ☐ Assignment Papers (cover sheet & document(s)) Name of Assignee 11. ☐ 37 CFR 3.73(c) Statement (when there is an assignee) ☒ Power of Attorney 12. ☐ English Translation Document (if applicable) 13. ☒ Information Disclosure Statement (PTO/SB/08 or PTO-1449) ☐ Copies of citations attached 14. ☐ Preliminary Amendment 15. ☐ Return Receipt Postcard (MPEP § 503) (Should be specifically itemized) 16. ☐ Certified Copy of Priority Document(s) (if foreign priority is claimed) 17. ☐ Nonpublication Request Under 35 U.S.C. 122(b)(2)(B)(i). Applicant must attach form PTO/SB/35 or equivalent. 18. ☒ Other: Communication Regarding Certified Copy *Note: (1) Benefit claims under 37 CFR 1.78 and foreign priority claims under 1.55 must be included in an Application Data Sheet (ADS). (2) For applications filed under 35 U.S.C. 111, the application must contain an ADS specifying the applicant if the applicant is an assignee, person to whom the inventor is under an obligation to assign, or person who otherwise shows sufficient proprietary interest in the matter. See 37 CFR 1.46(b). 19. CORRESPONDENCE ADDRESS ☒ The address associated with Customer Number: 86765 OR ☐ Correspondence address below <table><tr><td>Name</td><td colspan="3"></td></tr><tr><td>Address</td><td colspan="3"></td></tr><tr><td>City</td><td>State</td><td>Zip Code</td><td></td></tr><tr><td>Country</td><td>Telephone</td><td>Email</td><td></td></tr><tr><td>Signature</td><td>/Joseph H. Karlin/</td><td>Date</td><td>May 14, 2025</td></tr><tr><td>Name (Print/Type)</td><td>Joseph H. Karlin</td><td>Registration No. (Attorney/Agent)</td><td>62,471</td></tr></table>	Name				Address				City	State	Zip Code		Country	Telephone	Email		Signature	/Joseph H. Karlin/	Date	May 14, 2025	Name (Print/Type)	Joseph H. Karlin	Registration No. (Attorney/Agent)	62,471
Name																								
Address																								
City	State	Zip Code																						
Country	Telephone	Email																						
Signature	/Joseph H. Karlin/	Date	May 14, 2025																					
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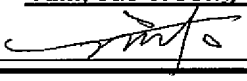
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DECLARATION AND ASSIGNMENT FOR UTILITY OR DESIGN APPLICATIONS		Attorney Docket Number:	
Title of Invention	SECONDARY BATTERY AND METHOD FOR MANUFACTURING THE SAME		
As the below named inventor(s) (hereinafter designated as the undersigned), I/We hereby declare that: This declaration is directed to: ☒ The attached application, or ☐ United States application or PCT international application number _____ filed on _____ The above-identified application was made or authorized to be made by me/us. I/We believe that I am the original inventor or an original joint inventor of a claimed invention in the application. I/We hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both. WHEREAS, the undersigned as a sole inventor (if only one name is listed below) or as a joint inventor (if plural names are listed below) has invented certain new and useful improvements in the application identified above. AND, WHEREAS LG CHEM, LTD., (hereinafter referred to as assignee), a corporation of Korea whose mailing address is 128, Yeoui-daero, Yeongdeungpo-gu, Seoul 07336, Republic of Korea is desirous of acquiring the entire right, title and interest in the same. NOW, THEREFORE, in consideration of the sum of one dollar, the receipt whereof is acknowledged, and other good and valuable consideration, the undersigned hereby sell(s), assign(s) and set(s) over to said assignee my/our entire right, title and interest for the United States of America and all other countries in and to said invention and the aforesaid utility patent application and all original, divisional, continuation, substitute or reissue applications and patents applied for or granted therefor in the United States of America and all other countries, for said invention, including without limitation all applications and patents for said invention claiming priority or benefit of the aforesaid utility application pursuant to any law or treaty, and including all claims for damages by reason of past infringement with the right to sue for and collect the same for its own use and the use of its successors, assigns and legal representatives, and including the right to claim such priority or benefit and the Commissioner of Patents and Trademarks is hereby authorized and requested to issue all patents on said improvements or resulting therefrom to said assignee herein, as assignee of the entire interest therein; and the undersigned for me/us and my/our legal representatives, heirs and assigns do hereby agree and covenant without further remuneration, to execute and deliver all original, divisional, continuation, reissue and other applications for Letters Patent on said improvements and all assignments thereof to said assignee or its assigns, to communicate to said assignee or its representatives all facts known to the undersigned respecting said improvements, whenever requested, to testify in any interferences or other legal proceedings in which any of said applications or patents may become involved, to sign all lawful papers, make all rightful oaths, and to do generally everything necessary to aid assignee, its successors, assigns and nominees to obtain patent protection for said improvements in all countries, the expenses incident to said applications to be borne and paid by said assignee. And I do hereby authorize my attorneys to insert on this deed the filing date and application number of said application when known.			
FULL NAME OF INVENTOR(S)			
Inventor one: <u>KIM, Jae Woong</u>		Date: <u>September 25, 2019</u>	
Signature: <u></u>			
☒ Additional inventors or a legal representative are being named on <u>1</u> additional form(s) attached hereto.			
PLEASE SEE THE WARNINGS ON THE LAST PAGE OF THIS FORM.			



SUPPLEMENTAL SHEET

DECLARATION AND ASSIGNMENT FOR UTILITY OR DESIGN APPLICATIONS	ADDITIONAL INVENTOR(S) Supplemental Sheet
Inventor two: <u>KIM, Hyoung Kwon</u> Signature: <u>[Signature]</u>	Date: <u>September 25, 2019</u>
Inventor three: <u>KIM, Sang Uck</u> Signature: <u>[Signature]</u>	Date: <u>September 25, 2019</u>
Inventor four: _____ Signature: _____	Date: _____
Inventor five: _____ Signature: _____	Date: _____
Inventor six: _____ Signature: _____	Date: _____
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SECONDARY BATTERY AND METHOD FOR MANUFACTURING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. Patent Application No. 18/620,269, filed on March 28, 2024, which is a continuation of U.S. Patent Application No. 18/618,609, filed on March 27, 2024, which is a continuation of U.S. Application No. 18/132,719, filed on April 10, 2023, now U.S. Patent No. 11,973,177, which is a continuation of U.S. Application No. 16/498,751, filed on September 27, 2019, now U.S. Patent No. 11,652,232, which is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/KR2018/004366, filed on April 13, 2018, which claims priority from Korean Patent Application Nos. 10-2017-0048645, filed on April 14, 2017, and 10-2018-0043191, filed on April 13, 2018, the disclosures of all of which are hereby incorporated by reference herein in their entireties.

TECHNICAL FIELD

[0002] The present invention relates to a secondary battery and a method for manufacturing the same, and more particularly, to a secondary battery in which an electrode tab increases in mechanical strength and a method for manufacturing the same.

BACKGROUND ART

[0003] Batteries (cells) that generate electric energy through physical or chemical reaction to supply the generated electric energy to the outside are used when AC power to be supplied to the building is not obtained, or DC power is required according to the living environments surrounded by various electric and electronic devices.

[0004] Among such batteries, primary batteries and secondary batteries, which are chemical batteries using chemical reaction, are being generally used. The primary batteries are consumable batteries which are collectively referred to as dry batteries. Also,

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secondary batteries are rechargeable batteries that are manufactured by using a material in a redox process between current and a substance is repeatable several times. When the reduction reaction is performed on the material by the current, power is charged, and when the oxidation reaction is performed on the material, power is discharged. Such the charging-discharging is repeatedly performed to generate electricity.

[0005] A lithium ion battery of the secondary batteries is manufactured through the following processes. An active material is applied to each of a positive electrode conductive foil and a negative electrode conductive foil at a predetermined thickness, and a separator is disposed between the positive electrode conductive foil and the negative electrode conductive foil, and then, an electrode assembly, in which the positive electrode conductive foil, the separator, and the negative electrode conductive foil are wound several times in a jelly-roll or cylindrical shape, is accommodated into a cylindrical or prismatic can, a pouch, and the like to seal the resultant product, thereby manufacturing the lithium ion battery.

[0006] An electrode assembly and a secondary battery including the same according to the related art is disclosed in Korea Patent Publication No. 10-2012-0006389.

[0007] In the secondary battery according to the related art, it is necessary to apply a technology in which a metal having low resistance is applied to a negative electrode tab, or the number of electrode tabs increases in order to develop a low resistance cell.

[0008] A negative multi-tab structure developed by this necessity was a two-tab structure in which an electrode tab is attached to each of a core and an outer portion or a three-tab structure in which an electrode tab is applied to each of a core and an outer portion, and a middle tab is attached to an intermediate portion.

[0009] However, in order to attach the middle tab on the characteristics of the jelly roll, a non-coating portion has to be formed on a predetermined portion in the middle of the

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electrode. Thus, there is a problem that the battery capacity is lowered due to an increase of the non-coating portion.

DISCLOSURE OF THE INVENTION

TECHNICAL PROBLEM

[0010] Therefore, the present invention has been made to solve the abovementioned problem, and an object of the present invention is to provide a secondary battery in which a multi-tab is formed while minimizing a non-coating portion and a method for manufacturing the same.

[0011] Also, an object of the present invention is to provide a secondary battery which is capable of increasing in mechanical strength while notching electrode foil to form an electrode tab and a method for manufacturing the same.

TECHNICAL SOLUTION

[0012] A secondary battery according to an embodiment of the present invention includes: an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; and a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state.

[0013] The notching tab part may continuously extend across 1/3 or more of a total length of the coating portion in the longitudinal direction.

[0014] When the electrode is in the wound state, the notching tab part may form a plurality of notching tabs.

[0015] The foil tab may have a width of 4 mm to 120 mm.

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[0016] The electrode may be accommodated in a can member, and each of the plurality of notching tabs may be in a bent state and may be connected to the can member.

[0017] A bent part formed by bending each of the plurality of notching tabs may be bonded to the can member.

[0018] The secondary battery may further include an insulation member disposed between the bent parts and an end of the electrode.

[0019] The secondary battery may further include an electrode tab attached to the non-coating portion.

[0020] When the electrode is in the wound state, the notching tab part may form a plurality of notching tabs, and the electrode may be accommodated in a can member, and each of the plurality of notching tabs may be in a bent state and may overlap the electrode tab.

[0021] The electrode tab and the plurality of notching tabs may be bonded together to the can member.

[0022] A method for manufacturing a secondary battery according to an embodiment of the present invention includes: a preparation step of preparing an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; a notching tab part formation step of notching the electrode to form a notching tab part not coated with the active material and extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction, the notching tab part extending across 1/3 or more of a total length of the coating portion in the longitudinal direction; a winding step including stacking the electrode with a separator and with another electrode having a polarity different from that of the electrode and winding the electrode, the separator, and the another electrode to form an

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electrode assembly; a notching tab formation step of cutting the notching tab part of the electrode assembly to form a plurality of notching tabs; and an installation step of accommodating the electrode assembly into a can member and bonding each of the plurality of notching tabs to the can member.

[0023] The electrode may be a negative electrode.

[0024] During the preparation step, the coating portion coated with the active material on the electrode collector and the non-coating portion on which the active material is not applied may be arranged in parallel to each other in the width direction of the electrode collector.

[0025] during the notching tab part formation step, the notching tab part on which the active material is not applied may be notched in the width direction of the coating portion while cutting the coating portion and the non-coating portion in the longitudinal direction of the electrode collector.

[0026] During the notching tab part formation step, notching may be performed so that the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the electrode.

[0027] During the notching tab formation step, the notching tab part may be cut using a laser.

[0028] During the installation step, the electrode assembly may be accommodated in the can member in a state in which the plurality of notching tabs are bent and bonded to the can member.

[0029] During the installation step, a bent portion of each of the plurality of notching tabs may be welded to the can member.

[0030] During the installation step, bent portions of each of the plurality of notching tabs may overlap each other and may be bonded to the can member together.

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[0031] During the winding step, the notching tab part may be wound to form two or more overlapping layers.

ADVANTAGEOUS EFFECTS

[0032] According to the present invention, the multi-tab may be formed while minimizing the formation of the non-coating portion.

[0033] According to the present invention, the multi-tab may be formed to minimize the electric resistance.

[0034] According to the present invention, the formation of the non-coating part may be minimized to secure the battery capacity.

[0035] According to the present invention, the notching tab may overlap each other to form two or more layers while forming the notching tab that is the foil tab from the electrode foil, whereby increasing in mechanical strength.

[0036] According to the present invention, the notching tab that is the foil tab may increase in mechanical strength to prevent the notching tab from being broken in the bent state, and thus, the notching tab may be bonded to the can member in the bent state to secure the reliability in electrical connection and improve the bonding performance.

[0037] According to the present invention, the notching tab and the electrode tab may be bonded to the can member at the same to secure the welding processability.

[0038] According to the present invention, the plurality of notching tabs may be formed outside the electrode assembly to install the insulation member between the plurality of notching tabs and the end of the electrode assembly, and thereby to prevent the notching tabs and the electrode assembly from coming into contact with each other, thereby improving the safety.

BRIEF DESCRIPTION OF THE DRAWINGS

[0039] FIG. 1 is a development view of an electrode according to an embodiment of the present invention.

[0040] FIG. 2 is a perspective view illustrating a state in which the electrode of FIG. 1 is wound.

[0041] FIG. 3 is a perspective view illustrating a state in which a notching tab part is cut to form a notching tab in FIG. 2.

[0042] FIG. 4 is a projective perspective view when seeing through a can member in a state in which an electrode assembly is installed in the can member according to an embodiment of the present invention.

[0043] FIG. 5 is a bottom view of only the electrode assembly in FIG. 4.

[0044] FIG. 6 is a flowchart successively illustrating a method for manufacturing a secondary battery according to an embodiment of the present invention.

[0045] FIG. 7 is a plan view illustrating a state in which a coating portion and a non-coating portion, which are formed in a longitudinal direction of an electrode collector, are arranged in parallel to each other in a width direction of the electrode collector according to an embodiment of the present invention.

[0046] FIG. 8 is a bottom view illustrating a state in which a notching tab and an electrode tab are bonded to each other according to another embodiment of the present invention.

[0047] FIG. 9 is a side view of FIG. 8.

MODE FOR CARRYING OUT THE INVENTION

[0048] Hereinafter, a secondary battery and a method for manufacturing the same according to exemplary embodiments of the present invention will be described in detail with reference to the accompanying drawings.

[0049] Terms or words used in the specification and claims should not be construed as limited to a lexical meaning, and should be understood as appropriate notions by the inventor based on that he/she is able to define terms to describe his/her invention in the best way to be seen by others. Therefore, embodiments and drawings described herein are simply exemplary and not exhaustive, and it will be understood that various equivalents may be made to take the place of the embodiments.

[0050] In the drawings, the dimension of each of components or a specific portion constituting the component is exaggerated, omitted, or schematically illustrated for convenience in description and clarity. Thus, the dimension of each element does not entirely reflect an actual size. Moreover, detailed descriptions related to well-known functions or configurations will be ruled out in order not to unnecessarily obscure subject matters of the present invention.

[0051] FIG. 1 is a development view of an electrode according to an embodiment of the present invention, and FIG. 2 is a perspective view illustrating a state in which the electrode of FIG. 1 is wound.

[0052] As illustrated in FIGS. 1 and 2, a secondary battery according to an embodiment of the present invention includes an electrode on which a coating portion 110 coated with an active material a on an electrode collector and a non-coating portion 120, on which the active material a is not applied on the electrode collector, are disposed in a longitudinal direction of the electrode collector and a notching tab part 111 extending from the coating portion 110 in a width direction of the electrode collector without being coated with the active material a and overlapping each other to form two or more layers when the electrode 100 is wound (FIG. 2 illustrates a notching tab part 111 that overlap each other to form two or more layers).

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[0053] The foil tab may overlap each other to form two or more layers while forming the notching tab part that is a foil tab from electrode foil of the electrode collector, thereby increasing in mechanical strength.

[0054] The electrode 100 may be one of a positive electrode coated with a positive electrode active material and a negative electrode coated with a negative electrode active material. The electrode assembly 1 may be manufactured by stacking the positive electrode, the negative electrode, and a separator several times so that the separator is disposed between the positive electrode and the negative electrode.

[0055] Also, the electrode assembly 1 may be manufactured by winding a stack, in which the positive electrode, the separator, and the negative electrode are stacked, in a jelly-roll shape.

[0056] The positive electrode may be an aluminum electrode collector and include a positive electrode coating portion coated with the positive electrode active material and a positive electrode non-coating portion which is not coated with the positive electrode active material.

[0057] The positive electrode active material may include a lithium-containing transition metal oxide such as LiCoO_2 , LiNiO_2 , LiMnO_2 , and LiMnO_4 or a lithium chalcogenide compound.

[0058] The positive electrode coating portion may be manufactured, for example, by applying the positive electrode active material to a portion of at least one surface of the aluminum electrode collector, and the remaining portion of the aluminum electrode collector, which is not coated with the positive electrode active material, may be the positive electrode non-coating portion.

[0059] The negative electrode may be a copper electrode collector and include a negative electrode coating portion coated with the negative electrode active material and a

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negative electrode non-coating portion which is not coated with the negative electrode active material.

[0060] The negative electrode active material may be a carbon material such as crystalline carbon, amorphous carbon, a carbon composite, and a carbon fiber, a lithium metal, or a lithium alloy.

[0061] The negative electrode coating portion may be manufactured, for example, by applying the negative electrode active material to a portion of at least one surface of the copper electrode collector, and the remaining portion of the copper electrode collector, which is not coated with the negative electrode active material, may be the negative electrode non-coating portion.

[0062] The separator may be manufactured by applying a polyvinylidene fluoride-hexafluoropropylene copolymer (PVDF-HFP co-polymer) to one base material selected from the group consisting of, for example, polyethylene (PE), polystyrene (PS), polypropylene (PP), and a copolymer of polyethylene (PE) and polypropylene.

[0063] As described above, the electrode 100 according to the present invention may be one of the positive electrode and the negative electrode, which constitute the electrode assembly 1. Particularly, the electrode 100 may be one, which requires further reduction in resistance, of the positive electrode and the negative electrode.

[0064] In the present invention, the negative electrode will be described as an example.

[0065] The electrode 100 has one end on which the non-coating portion 120 is formed. An electrode tab 121 may be attached to the non-coating portion 120 formed on the one end of the electrode 100.

[0066] The notching tab part 111 may be formed by notching the electrode 100 to extend from the coating portion 110 of the electrode 100 in the width direction. Only the electrode

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collector may be formed to extend on an area corresponding to 1/3 or more of the total length of the coating portion 110 of the electrode 100, and the active material a may not be applied.

[0067] Only the active material a may be removed from the coating portion 110 extending from the electrode 100 to form the notching tab part 111. Alternatively, only the electrode foil may extend without applying the active material a on the area of the coating portion 110 of the electrode 100 to form the notching tab part 111.

[0068] Thus, there may be an effect of forming the multi-tab while minimizing the formation of the non-coating portion, and the formation of the non-coating portion may be minimized to secure the battery capacity.

[0069] FIG. 3 is a perspective view illustrating a state in which the notching tab part is cut to form the notching tab in FIG. 2.

[0070] As illustrated in FIG. 3, the notching tab part 111 may be cut in the wound state to form a plurality of notching tabs 111a. When the number of notching tabs 111a increases, the electrode tab may decrease in resistance.

[0071] Here, one notching tab 111a may have a width of 4 mm to 5 mm. This is done for a reason in which, if the notching tab 111a has a width that is less than 4 mm, the mechanical strength may be weakened, and if the notching tab 111a has a width exceeding 5 mm, deformation in the process of bonding for the electrical connection with the can member may be difficult.

[0072] The notching tab 111a that is the foil tab may increase in mechanical strength to prevent the notching tab 111a from being broken in the bent state, and thus, the notching tab 111a may be bonded to the can member in the bent state to secure the reliability in electrical connection and improve the bonding performance.

[0073] FIG. 4 is a projective perspective view when seeing through the can member in a state in which the electrode assembly is installed in the can member according to an

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embodiment of the present invention, and FIG. 5 is a bottom view of only the electrode assembly in FIG. 4.

[0074] As illustrated in FIGS. 4 and 5, in the electrode assembly 1 of the secondary battery according to the present invention, a bent part of the notching tab 111a may be welded to be coupled to the can member 10 so that the notching tab 111a is electrically connected to the can member 10 in a state in which the notching tab 111a is bent while being accommodated in the can member 10.

[0075] The notching tab 111a may be bent to allow an end of the notching tab 111a to be coupled to a center of a bottom surface of the can member 10. Thus, the plurality of notching tabs 111a may be electrically connected to each other while the electrical connection is smooth to minimize the electric resistance.

[0076] FIG. 8 is a bottom view illustrating a state in which a notching tab and an electrode tab are bonded to each other according to another embodiment of the present invention, and FIG. 9 is a side view of FIG. 8.

[0077] As illustrated in FIGS. 8 and 9, in a secondary battery according to another embodiment of the present invention, a notching tab part 111 of an electrode assembly 1, in which an electrode 100 is wound, may be cut to form a plurality of notching tabs 111a. The plurality of notching tabs 111a may be bent and then be connected to overlap an electrode tab 121 formed on a central portion of an end of the electrode assembly 1.

[0078] Also, the electrode assembly 1 may be accommodated in a can member 10, and the plurality of notching tabs 111a connected to overlap the electrode tab 121 may be bonded to the can member 10 together with the electrode tab 121 through welding.

[0079] As described above, in the secondary battery according to another embodiment of the present invention, since the plurality of notching tabs 111a and the electrode tab 121 are welded together to the can member 10 in the state in which the notching tabs 111a and

the electrode tab 121 are connected to overlap each other, welding processability may be easily secured.

[0080] In the secondary battery according to another embodiment of the present invention, when an end of the electrode assembly 1 has a length ranging of 16 mm to 21 mm, a notching length of each of the plurality of notching tabs 111a may range of 8 mm to 11 mm. When the notching length of the notching tab 111a is less than 8 mm, the length of the notching tab 111a may be shortened, and the notching tab 111a may be bent. Thus, it may be difficult to allow the notching tab 111a to overlap the electrode tab 121 formed on a central portion of an end of the electrode assembly 1. Also, when the notching length of the notching tab 111a exceeds 11 mm, the length of the notching tab 111a may be unnecessarily lengthened to increase in cost due to waste of the material and in space occupied by the notching tab 111a within the can member 10. Thus, an amount of electrolyte filled into the can member 10 may be reduced to deteriorate capacity. Preferably, the notching tab 111a may have a notching length of 9 mm to 10 mm.

[0081] In the secondary battery according to another embodiment of the present invention, an insulation member 130 may be formed between a bent part of the notching tab 111a and an end of the electrode assembly 1 in which the electrode 100 is wound.

[0082] The insulation member 130 may insulate the notching tab 111a from the end of the electrode assembly 1 to prevent the notching tab 111a from coming into contact with the electrode assembly 1.

[0083] The plurality of notching tabs 111a may be formed on an outer portion of the end of the electrode assembly to prevent the insulation member 130 from interfering with the end of the electrode assembly 1.

[0084] The plurality of notching tabs 111a may be formed on the outside of the electrode assembly so that the insulation member 130 is installed between the plurality of notching

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tabs 111a and the end of the electrode assembly 1. Thus, the contact between the notching tab 111a and the electrode assembly 1 may be prevented to improve safety.

[0085] Hereinafter, a method of manufacturing a secondary battery according an embodiment of the present invention will be described in detail with reference to the accompanying drawings.

[0086] FIG. 6 is a flowchart successively illustrating a method for manufacturing a secondary battery according to an embodiment of the present invention.

[0087] As illustrated in FIG. 6, a method for manufacturing a secondary battery according to an embodiment of the present invention includes a preparation step (S1), a notching tab part formation step (S2), a winding step (s3), a notching tab formation step (S4), and an installation step (S5).

[0088] The preparation step (S1) is a step of preparing an electrode 100 in which a coating portion 110 coated with an active material a on an electrode collector and a non-coating portion 120 on which the active material a is not applied are provided in a longitudinal direction of the electrode collector.

[0089] Here, the electrode 100 may be a negative electrode.

[0090] FIG. 7 is a plan view illustrating a state in which the coating portion and the non-coating portion, which are formed in the longitudinal direction of the electrode collector, are arranged in parallel to each other in a width direction of the electrode collector according to an embodiment of the present invention.

[0091] As illustrated in FIG. 7, in the preparation step (S1), the coating portion 110 and the non-coating portion 120, which are formed in a longitudinal direction L of the electrode collector may be arranged in parallel to each other in the width direction W of the electrode collector.

[0092] In the notching tab part formation step (S2) is a step of notching the electrode 10 to form a notching tab part 111 extending in a state in which the active material is not formed in the width direction W of the coating portion 111 on an area of the electrode 100 corresponding to 1/3 or more of a length of the coating portion 110.

[0093] As described above, the notching of the notching tab part 111 in the electrode 100 may be a step of notching a portion of the non-coating portion on which the active material is not applied and which is formed in the width direction W of the coating portion 11 while cutting the electrode 100 in the electrode collector to form a notch tab part 11.

[0094] Also, when the coating portion 100 and the non-coating portion 120, which are formed in the longitudinal direction L of the electrode collector, are arranged in parallel to each other in the width direction W of the electrode collector, the non-coating portion on which the active material is not applied and which is formed between one coating portion 110 and the other coating portion 110a, which are arranged in parallel to each other in the width direction W may be notched while cutting the coating portion 110 and the non-coating portion 120, which are arranged in parallel to each other to form the notching tab part 111 in the width direction W of the coating portion 110.

[0095] The winding step (S3) may be a step of stacking the electrode 100, on which the notching tab part 111 is formed, on the separator and the other electrode having a polarity different from that of the electrode 100 and winding the stack to form the electrode assembly 1.

[0096] While the electrode 100 is wound, the notching tab part 111 may overlap each other to form two or more layers.

[0097] The notching tab formation step (S4) may be a step of cutting the notching tab part 111 to form a plurality of notching tabs 111a.

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[0098] The notching tabs 111a formed by cutting the notching tab part 111 may overlap each other to two or more layers like the notching tab part 111.

[0099] Here, the cutting of the notching tab part 111 may be maximized in work efficiency by using laser.

[0100] The insulation step (S5) may be a step of accommodating the electrode assembly 1 into the can member 10 to bond the notching tabs 111a to the can member 10.

[0101] In the installation step (S5), each of the notching tab 111a may be bent so that the welding is easily performed while the notching tab 111a is electrically connected to the can member 10 to allow the electrode assembly 1 to be accommodated in the can member 10 in the state in which the notching tab 111a is bent.

[0102] Here, since the notching tab 111a has the two or more layers, even though the notching tab 111a is bent, mechanical strength may be maintained, and thus, the notching tab may not be easily broken.

[0103] The bent notching tab 111a may have an end that faces a center of a bottom surface of the can member 10.

[0104] The coupling between the bottom surface of the can member 10 and the notching tab 111a may be performed by welding the bent part of the notching tab 111a to the bottom surface of the can member 10.

[0105] In addition, according to another embodiment of the present invention, the bent portions of the plurality of notching tabs 111a may overlap the central portion of the end of the electrode assembly 1 so as to be welded and bonded to the can member 10 together with the electrode tab 121 formed on the central portion of the end of the electrode 1 at the same time.

[0106] Although the secondary battery and the method for manufacturing the same according to the present invention has been described above with reference to the exemplary

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drawings, various changes and modifications may be made thereto by one skilled in the art without departing from the scope and spirit of the invention as set forth in the appended claims.

CLAIMS

1. A secondary battery, comprising:
a can member; and
an electrode assembly having a positive electrode, a negative electrode and a separator disposed therebetween, the positive electrode, the negative electrode, and the separator being wound together about an axis into a jelly-roll shape,
wherein the positive electrode or the negative electrode comprises:
an electrode collector;
a coating portion having an active material coated on the electrode collector;
a first non-coating portion on which the active material is not applied to the electrode collector, the first non-coating portion being adjacent to the coating portion in a width direction perpendicular to a longitudinal direction of the electrode collector, the first non-coating portion having a longitudinal dimension extending along the longitudinal direction of the electrode collector, the first non-coating portion configured to be bent and electrically coupled to the can member; and
a second non-coating portion on which the active material is not applied to the electrode collector, the second non-coating portion being adjacent to the coating portion in the longitudinal direction.
2. The secondary battery of claim 1, wherein the second non-coating portion is located at a longitudinal end of the electrode collector.
3. The secondary battery of claim 1, wherein the first non-coating portion is spaced apart from the second non-coating portion.
4. The secondary battery of claim 3, wherein the first non-coating portion is spaced apart from the second non-coating portion along the longitudinal direction of the electrode collector.
5. The secondary battery of claim 1, wherein the electrode collector defines a first width in the width direction along the first non-coating portion in the longitudinal

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direction, and wherein the electrode collector defines a second width in the width direction along a segment of the electrode collector in the longitudinal direction that excludes the first non-coating portion, the first width being greater than the second width.

6. The secondary battery of claim 5, wherein the segment of the electrode collector is a remainder of the electrode collector in the longitudinal direction that excludes the first non-coating portion.

7. The secondary battery of claim 5, wherein the segment of the electrode collector extends along the second non-coating portion in the longitudinal direction.

8. The secondary battery of claim 1, wherein the first non-coating portion continuously extends across $\frac{1}{3}$ or more of a total length of the coating portion in the longitudinal direction.

9. The secondary battery of claim 1, wherein the first non-coating portion continuously extends across at least two circumferential windings of the electrode collector in the longitudinal direction.

10. The secondary battery of claim 1, wherein the first non-coating portion is attached to the can member.

11. The secondary battery of claim 1, further comprising an electrode tab attached to and extending from the second non-coating portion.

12. The secondary battery of claim 1, wherein the first non-coating portion is configured such that an electric current in the first non-coating portion flows toward the can member in the width direction.

13. The secondary battery of claim 1, wherein the first non-coating portion is formed from a metal foil of the electrode collector.

14. The secondary battery of claim 1, wherein the can member has a cylindrical shape.

ABSTRACT

An electrode for a secondary battery may include an electrode collector, a coating portion having an active material coated on the electrode collector, a first non-coating portion on which the active material is not applied to the electrode collector, and a second non-coating portion on which the active material is not applied to the electrode collector. The first non-coating portion may be adjacent to the coating portion in a width direction perpendicular to a longitudinal direction of the electrode collector, the first non-coating portion having a longitudinal dimension extending along the longitudinal direction, the first non-coating portion configured to be bent and electrically coupled to a can member of the secondary battery. The second non-coating portion may be adjacent to the coating portion in the longitudinal direction. The electrode may be stacked into a stack with a separator and a second electrode of an opposite polarity than the electrode.

SECONDARY BATTERY AND METHOD FOR MANUFACTURING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. Patent Application No. 18/620,269, filed on March 28, 2024, which is a continuation of U.S. Patent Application No. 18/618,609, filed on March 27, 2024, which is a continuation of U.S. Application No. 18/132,719, filed on April 10, 2023, now U.S. Patent No. 11,973,177, which is a continuation of U.S. Application No. 16/498,751, filed on September 27, 2019, now U.S. Patent No. 11,652,232, which is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/KR2018/004366, filed on April 13, 2018, which claims priority from Korean Patent Application Nos. 10-2017-0048645, filed on April 14, 2017, and 10-2018-0043191, filed on April 13, 2018, the disclosures of all of which are hereby incorporated by reference herein in their entireties.

TECHNICAL FIELD

[0002] The present invention relates to a secondary battery and a method for manufacturing the same, and more particularly, to a secondary battery in which an electrode tab increases in mechanical strength and a method for manufacturing the same.

BACKGROUND ART

[0003] Batteries (cells) that generate electric energy through physical or chemical reaction to supply the generated electric energy to the outside are used when AC power to be supplied to the building is not obtained, or DC power is required according to the living environments surrounded by various electric and electronic devices.

[0004] Among such batteries, primary batteries and secondary batteries, which are chemical batteries using chemical reaction, are being generally used. The primary batteries are consumable batteries which are collectively referred to as dry batteries. Also,

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secondary batteries are rechargeable batteries that are manufactured by using a material in a redox process between current and a substance is repeatable several times. When the reduction reaction is performed on the material by the current, power is charged, and when the oxidation reaction is performed on the material, power is discharged. Such the charging-discharging is repeatedly performed to generate electricity.

[0005] A lithium ion battery of the secondary batteries is manufactured through the following processes. An active material is applied to each of a positive electrode conductive foil and a negative electrode conductive foil at a predetermined thickness, and a separator is disposed between the positive electrode conductive foil and the negative electrode conductive foil, and then, an electrode assembly, in which the positive electrode conductive foil, the separator, and the negative electrode conductive foil are wound several times in a jelly-roll or cylindrical shape, is accommodated into a cylindrical or prismatic can, a pouch, and the like to seal the resultant product, thereby manufacturing the lithium ion battery.

[0006] An electrode assembly and a secondary battery including the same according to the related art is disclosed in Korea Patent Publication No. 10-2012-0006389.

[0007] In the secondary battery according to the related art, it is necessary to apply a technology in which a metal having low resistance is applied to a negative electrode tab, or the number of electrode tabs increases in order to develop a low resistance cell.

[0008] A negative multi-tab structure developed by this necessity was a two-tab structure in which an electrode tab is attached to each of a core and an outer portion or a three-tab structure in which an electrode tab is applied to each of a core and an outer portion, and a middle tab is attached to an intermediate portion.

[0009] However, in order to attach the middle tab on the characteristics of the jelly roll, a non-coating portion has to be formed on a predetermined portion in the middle of the

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electrode. Thus, there is a problem that the battery capacity is lowered due to an increase of the non-coating portion.

DISCLOSURE OF THE INVENTION

TECHNICAL PROBLEM

[0010] Therefore, the present invention has been made to solve the abovementioned problem, and an object of the present invention is to provide a secondary battery in which a multi-tab is formed while minimizing a non-coating portion and a method for manufacturing the same.

[0011] Also, an object of the present invention is to provide a secondary battery which is capable of increasing in mechanical strength while notching electrode foil to form an electrode tab and a method for manufacturing the same.

TECHNICAL SOLUTION

[0012] A secondary battery according to an embodiment of the present invention includes: an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; and a notching tab part extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction and not coated with the active material, the notching tab part having a wound state including two or more overlapping layers when the electrode is in a wound state.

[0013] The notching tab part may continuously extend across 1/3 or more of a total length of the coating portion in the longitudinal direction.

[0014] When the electrode is in the wound state, the notching tab part may form a plurality of notching tabs.

[0015] The foil tab may have a width of 4 mm to 120 mm.

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[0016] The electrode may be accommodated in a can member, and each of the plurality of notching tabs may be in a bent state and may be connected to the can member.

[0017] A bent part formed by bending each of the plurality of notching tabs may be bonded to the can member.

[0018] The secondary battery may further include an insulation member disposed between the bent parts and an end of the electrode.

[0019] The secondary battery may further include an electrode tab attached to the non-coating portion.

[0020] When the electrode is in the wound state, the notching tab part may form a plurality of notching tabs, and the electrode may be accommodated in a can member, and each of the plurality of notching tabs may be in a bent state and may overlap the electrode tab.

[0021] The electrode tab and the plurality of notching tabs may be bonded together to the can member.

[0022] A method for manufacturing a secondary battery according to an embodiment of the present invention includes: a preparation step of preparing an electrode provided with a coating portion coated with an active material on an electrode collector and a non-coating portion on which the active material is not applied to the electrode collector in a longitudinal direction of the electrode collector; a notching tab part formation step of notching the electrode to form a notching tab part not coated with the active material and extending from the coating portion in a width direction of the electrode collector perpendicular to the longitudinal direction, the notching tab part extending across 1/3 or more of a total length of the coating portion in the longitudinal direction; a winding step including stacking the electrode with a separator and with another electrode having a polarity different from that of the electrode and winding the electrode, the separator, and the another electrode to form an

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electrode assembly; a notching tab formation step of cutting the notching tab part of the electrode assembly to form a plurality of notching tabs; and an installation step of accommodating the electrode assembly into a can member and bonding each of the plurality of notching tabs to the can member.

[0023] The electrode may be a negative electrode.

[0024] During the preparation step, the coating portion coated with the active material on the electrode collector and the non-coating portion on which the active material is not applied may be arranged in parallel to each other in the width direction of the electrode collector.

[0025] during the notching tab part formation step, the notching tab part on which the active material is not applied may be notched in the width direction of the coating portion while cutting the coating portion and the non-coating portion in the longitudinal direction of the electrode collector.

[0026] During the notching tab part formation step, notching may be performed so that the notching tab part on which the active material is not applied is notched in the width direction of the coating portion while cutting the electrode.

[0027] During the notching tab formation step, the notching tab part may be cut using a laser.

[0028] During the installation step, the electrode assembly may be accommodated in the can member in a state in which the plurality of notching tabs are bent and bonded to the can member.

[0029] During the installation step, a bent portion of each of the plurality of notching tabs may be welded to the can member.

[0030] During the installation step, bent portions of each of the plurality of notching tabs may overlap each other and may be bonded to the can member together.

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[0031] During the winding step, the notching tab part may be wound to form two or more overlapping layers.

ADVANTAGEOUS EFFECTS

[0032] According to the present invention, the multi-tab may be formed while minimizing the formation of the non-coating portion.

[0033] According to the present invention, the multi-tab may be formed to minimize the electric resistance.

[0034] According to the present invention, the formation of the non-coating part may be minimized to secure the battery capacity.

[0035] According to the present invention, the notching tab may overlap each other to form two or more layers while forming the notching tab that is the foil tab from the electrode foil, whereby increasing in mechanical strength.

[0036] According to the present invention, the notching tab that is the foil tab may increase in mechanical strength to prevent the notching tab from being broken in the bent state, and thus, the notching tab may be bonded to the can member in the bent state to secure the reliability in electrical connection and improve the bonding performance.

[0037] According to the present invention, the notching tab and the electrode tab may be bonded to the can member at the same to secure the welding processability.

[0038] According to the present invention, the plurality of notching tabs may be formed outside the electrode assembly to install the insulation member between the plurality of notching tabs and the end of the electrode assembly, and thereby to prevent the notching tabs and the electrode assembly from coming into contact with each other, thereby improving the safety.

BRIEF DESCRIPTION OF THE DRAWINGS

[0039] FIG. 1 is a development view of an electrode according to an embodiment of the present invention.

[0040] FIG. 2 is a perspective view illustrating a state in which the electrode of FIG. 1 is wound.

[0041] FIG. 3 is a perspective view illustrating a state in which a notching tab part is cut to form a notching tab in FIG. 2.

[0042] FIG. 4 is a projective perspective view when seeing through a can member in a state in which an electrode assembly is installed in the can member according to an embodiment of the present invention.

[0043] FIG. 5 is a bottom view of only the electrode assembly in FIG. 4.

[0044] FIG. 6 is a flowchart successively illustrating a method for manufacturing a secondary battery according to an embodiment of the present invention.

[0045] FIG. 7 is a plan view illustrating a state in which a coating portion and a non-coating portion, which are formed in a longitudinal direction of an electrode collector, are arranged in parallel to each other in a width direction of the electrode collector according to an embodiment of the present invention.

[0046] FIG. 8 is a bottom view illustrating a state in which a notching tab and an electrode tab are bonded to each other according to another embodiment of the present invention.

[0047] FIG. 9 is a side view of FIG. 8.

MODE FOR CARRYING OUT THE INVENTION

[0048] Hereinafter, a secondary battery and a method for manufacturing the same according to exemplary embodiments of the present invention will be described in detail with reference to the accompanying drawings.

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[0049] Terms or words used in the specification and claims should not be construed as limited to a lexical meaning, and should be understood as appropriate notions by the inventor based on that he/she is able to define terms to describe his/her invention in the best way to be seen by others. Therefore, embodiments and drawings described herein are simply exemplary and not exhaustive, and it will be understood that various equivalents may be made to take the place of the embodiments.

[0050] In the drawings, the dimension of each of components or a specific portion constituting the component is exaggerated, omitted, or schematically illustrated for convenience in description and clarity. Thus, the dimension of each element does not entirely reflect an actual size. Moreover, detailed descriptions related to well-known functions or configurations will be ruled out in order not to unnecessarily obscure subject matters of the present invention.

[0051] FIG. 1 is a development view of an electrode according to an embodiment of the present invention, and FIG. 2 is a perspective view illustrating a state in which the electrode of FIG. 1 is wound.

[0052] As illustrated in FIGS. 1 and 2, a secondary battery according to an embodiment of the present invention includes an electrode on which a coating portion 110 coated with an active material a on an electrode collector and a non-coating portion 120, on which the active material a is not applied on the electrode collector, are disposed in a longitudinal direction of the electrode collector and a notching tab part 111 extending from the coating portion 110 in a width direction of the electrode collector without being coated with the active material a and overlapping each other to form two or more layers when the electrode 100 is wound (FIG. 2 illustrates a notching tab part 111 that overlap each other to form two or more layers).

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[0053] The foil tab may overlap each other to form two or more layers while forming the notching tab part that is a foil tab from electrode foil of the electrode collector, thereby increasing in mechanical strength.

[0054] The electrode 100 may be one of a positive electrode coated with a positive electrode active material and a negative electrode coated with a negative electrode active material. The electrode assembly 1 may be manufactured by stacking the positive electrode, the negative electrode, and a separator several times so that the separator is disposed between the positive electrode and the negative electrode.

[0055] Also, the electrode assembly 1 may be manufactured by winding a stack, in which the positive electrode, the separator, and the negative electrode are stacked, in a jelly-roll shape.

[0056] The positive electrode may be an aluminum electrode collector and include a positive electrode coating portion coated with the positive electrode active material and a positive electrode non-coating portion which is not coated with the positive electrode active material.

[0057] The positive electrode active material may include a lithium-containing transition metal oxide such as LiCoO_2 , LiNiO_2 , LiMnO_2 , and LiMnO_4 or a lithium chalcogenide compound.

[0058] The positive electrode coating portion may be manufactured, for example, by applying the positive electrode active material to a portion of at least one surface of the aluminum electrode collector, and the remaining portion of the aluminum electrode collector, which is not coated with the positive electrode active material, may be the positive electrode non-coating portion.

[0059] The negative electrode may be a copper electrode collector and include a negative electrode coating portion coated with the negative electrode active material and a

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negative electrode non-coating portion which is not coated with the negative electrode active material.

[0060] The negative electrode active material may be a carbon material such as crystalline carbon, amorphous carbon, a carbon composite, and a carbon fiber, a lithium metal, or a lithium alloy.

[0061] The negative electrode coating portion may be manufactured, for example, by applying the negative electrode active material to a portion of at least one surface of the copper electrode collector, and the remaining portion of the copper electrode collector, which is not coated with the negative electrode active material, may be the negative electrode non-coating portion.

[0062] The separator may be manufactured by applying a polyvinylidene fluoride-hexafluoropropylene copolymer (PVDF-HFP co-polymer) to one base material selected from the group consisting of, for example, polyethylene (PE), polystyrene (PS), polypropylene (PP), and a copolymer of polyethylene (PE) and polypropylene.

[0063] As described above, the electrode 100 according to the present invention may be one of the positive electrode and the negative electrode, which constitute the electrode assembly 1. Particularly, the electrode 100 may be one, which requires further reduction in resistance, of the positive electrode and the negative electrode.

[0064] In the present invention, the negative electrode will be described as an example.

[0065] The electrode 100 has one end on which the non-coating portion 120 is formed. An electrode tab 121 may be attached to the non-coating portion 120 formed on the one end of the electrode 100.

[0066] The notching tab part 111 may be formed by notching the electrode 100 to extend from the coating portion 110 of the electrode 100 in the width direction. Only the electrode

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collector may be formed to extend on an area corresponding to 1/3 or more of the total length of the coating portion 110 of the electrode 100, and the active material a may not be applied.

[0067] Only the active material a may be removed from the coating portion 110 extending from the electrode 100 to form the notching tab part 111. Alternatively, only the electrode foil may extend without applying the active material a on the area of the coating portion 110 of the electrode 100 to form the notching tab part 111.

[0068] Thus, there may be an effect of forming the multi-tab while minimizing the formation of the non-coating portion, and the formation of the non-coating portion may be minimized to secure the battery capacity.

[0069] FIG. 3 is a perspective view illustrating a state in which the notching tab part is cut to form the notching tab in FIG. 2.

[0070] As illustrated in FIG. 3, the notching tab part 111 may be cut in the wound state to form a plurality of notching tabs 111a. When the number of notching tabs 111a increases, the electrode tab may decrease in resistance.

[0071] Here, one notching tab 111a may have a width of 4 mm to 5 mm. This is done for a reason in which, if the notching tab 111a has a width that is less than 4 mm, the mechanical strength may be weakened, and if the notching tab 111a has a width exceeding 5 mm, deformation in the process of bonding for the electrical connection with the can member may be difficult.

[0072] The notching tab 111a that is the foil tab may increase in mechanical strength to prevent the notching tab 111a from being broken in the bent state, and thus, the notching tab 111a may be bonded to the can member in the bent state to secure the reliability in electrical connection and improve the bonding performance.

[0073] FIG. 4 is a projective perspective view when seeing through the can member in a state in which the electrode assembly is installed in the can member according to an

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embodiment of the present invention, and FIG. 5 is a bottom view of only the electrode assembly in FIG. 4.

[0074] As illustrated in FIGS. 4 and 5, in the electrode assembly 1 of the secondary battery according to the present invention, a bent part of the notching tab 111a may be welded to be coupled to the can member 10 so that the notching tab 111a is electrically connected to the can member 10 in a state in which the notching tab 111a is bent while being accommodated in the can member 10.

[0075] The notching tab 111a may be bent to allow an end of the notching tab 111a to be coupled to a center of a bottom surface of the can member 10. Thus, the plurality of notching tabs 111a may be electrically connected to each other while the electrical connection is smooth to minimize the electric resistance.

[0076] FIG. 8 is a bottom view illustrating a state in which a notching tab and an electrode tab are bonded to each other according to another embodiment of the present invention, and FIG. 9 is a side view of FIG. 8.

[0077] As illustrated in FIGS. 8 and 9, in a secondary battery according to another embodiment of the present invention, a notching tab part 111 of an electrode assembly 1, in which an electrode 100 is wound, may be cut to form a plurality of notching tabs 111a. The plurality of notching tabs 111a may be bent and then be connected to overlap an electrode tab 121 formed on a central portion of an end of the electrode assembly 1.

[0078] Also, the electrode assembly 1 may be accommodated in a can member 10, and the plurality of notching tabs 111a connected to overlap the electrode tab 121 may be bonded to the can member 10 together with the electrode tab 121 through welding.

[0079] As described above, in the secondary battery according to another embodiment of the present invention, since the plurality of notching tabs 111a and the electrode tab 121 are welded together to the can member 10 in the state in which the notching tabs 111a and

the electrode tab 121 are connected to overlap each other, welding processability may be easily secured.

[0080] In the secondary battery according to another embodiment of the present invention, when an end of the electrode assembly 1 has a length ranging of 16 mm to 21 mm, a notching length of each of the plurality of notching tabs 111a may range of 8 mm to 11 mm. When the notching length of the notching tab 111a is less than 8 mm, the length of the notching tab 111a may be shortened, and the notching tab 111a may be bent. Thus, it may be difficult to allow the notching tab 111a to overlap the electrode tab 121 formed on a central portion of an end of the electrode assembly 1. Also, when the notching length of the notching tab 111a exceeds 11 mm, the length of the notching tab 111a may be unnecessarily lengthened to increase in cost due to waste of the material and in space occupied by the notching tab 111a within the can member 10. Thus, an amount of electrolyte filled into the can member 10 may be reduced to deteriorate capacity. Preferably, the notching tab 111a may have a notching length of 9 mm to 10 mm.

[0081] In the secondary battery according to another embodiment of the present invention, an insulation member 130 may be formed between a bent part of the notching tab 111a and an end of the electrode assembly 1 in which the electrode 100 is wound.

[0082] The insulation member 130 may insulate the notching tab 111a from the end of the electrode assembly 1 to prevent the notching tab 111a from coming into contact with the electrode assembly 1.

[0083] The plurality of notching tabs 111a may be formed on an outer portion of the end of the electrode assembly to prevent the insulation member 130 from interfering with the end of the electrode assembly 1.

[0084] The plurality of notching tabs 111a may be formed on the outside of the electrode assembly so that the insulation member 130 is installed between the plurality of notching

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tabs 111a and the end of the electrode assembly 1. Thus, the contact between the notching tab 111a and the electrode assembly 1 may be prevented to improve safety.

[0085] Hereinafter, a method of manufacturing a secondary battery according an embodiment of the present invention will be described in detail with reference to the accompanying drawings.

[0086] FIG. 6 is a flowchart successively illustrating a method for manufacturing a secondary battery according to an embodiment of the present invention.

[0087] As illustrated in FIG. 6, a method for manufacturing a secondary battery according to an embodiment of the present invention includes a preparation step (S1), a notching tab part formation step (S2), a winding step (s3), a notching tab formation step (S4), and an installation step (S5).

[0088] The preparation step (S1) is a step of preparing an electrode 100 in which a coating portion 110 coated with an active material a on an electrode collector and a non-coating portion 120 on which the active material a is not applied are provided in a longitudinal direction of the electrode collector.

[0089] Here, the electrode 100 may be a negative electrode.

[0090] FIG. 7 is a plan view illustrating a state in which the coating portion and the non-coating portion, which are formed in the longitudinal direction of the electrode collector, are arranged in parallel to each other in a width direction of the electrode collector according to an embodiment of the present invention.

[0091] As illustrated in FIG. 7, in the preparation step (S1), the coating portion 110 and the non-coating portion 120, which are formed in a longitudinal direction L of the electrode collector may be arranged in parallel to each other in the width direction W of the electrode collector.

[0092] In the notching tab part formation step (S2) is a step of notching the electrode 10 to form a notching tab part 111 extending in a state in which the active material is not formed in the width direction W of the coating portion 111 on an area of the electrode 100 corresponding to 1/3 or more of a length of the coating portion 110.

[0093] As described above, the notching of the notching tab part 111 in the electrode 100 may be a step of notching a portion of the non-coating portion on which the active material is not applied and which is formed in the width direction W of the coating portion 11 while cutting the electrode 100 in the electrode collector to form a notch tab part 11.

[0094] Also, when the coating portion 100 and the non-coating portion 120, which are formed in the longitudinal direction L of the electrode collector, are arranged in parallel to each other in the width direction W of the electrode collector, the non-coating portion on which the active material is not applied and which is formed between one coating portion 110 and the other coating portion 110a, which are arranged in parallel to each other in the width direction W may be notched while cutting the coating portion 110 and the non-coating portion 120, which are arranged in parallel to each other to form the notching tab part 111 in the width direction W of the coating portion 110.

[0095] The winding step (S3) may be a step of stacking the electrode 100, on which the notching tab part 111 is formed, on the separator and the other electrode having a polarity different from that of the electrode 100 and winding the stack to form the electrode assembly 1.

[0096] While the electrode 100 is wound, the notching tab part 111 may overlap each other to form two or more layers.

[0097] The notching tab formation step (S4) may be a step of cutting the notching tab part 111 to form a plurality of notching tabs 111a.

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[0098] The notching tabs 111a formed by cutting the notching tab part 111 may overlap each other to two or more layers like the notching tab part 111.

[0099] Here, the cutting of the notching tab part 111 may be maximized in work efficiency by using laser.

[0100] The insulation step (S5) may be a step of accommodating the electrode assembly 1 into the can member 10 to bond the notching tabs 111a to the can member 10.

[0101] In the installation step (S5), each of the notching tab 111a may be bent so that the welding is easily performed while the notching tab 111a is electrically connected to the can member 10 to allow the electrode assembly 1 to be accommodated in the can member 10 in the state in which the notching tab 111a is bent.

[0102] Here, since the notching tab 111a has the two or more layers, even though the notching tab 111a is bent, mechanical strength may be maintained, and thus, the notching tab may not be easily broken.

[0103] The bent notching tab 111a may have an end that faces a center of a bottom surface of the can member 10.

[0104] The coupling between the bottom surface of the can member 10 and the notching tab 111a may be performed by welding the bent part of the notching tab 111a to the bottom surface of the can member 10.

[0105] In addition, according to another embodiment of the present invention, the bent portions of the plurality of notching tabs 111a may overlap the central portion of the end of the electrode assembly 1 so as to be welded and bonded to the can member 10 together with the electrode tab 121 formed on the central portion of the end of the electrode 1 at the same time.

[0106] Although the secondary battery and the method for manufacturing the same according to the present invention has been described above with reference to the exemplary

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drawings, various changes and modifications may be made thereto by one skilled in the art without departing from the scope and spirit of the invention as set forth in the appended claims.

CLAIMS

1. A secondary battery, comprising:
a can member; and
an electrode assembly having a positive electrode, a negative electrode and a separator disposed therebetween, the positive electrode, the negative electrode, and the separator being wound together about an axis into a jelly-roll shape,
wherein the positive electrode or the negative electrode comprises:
an electrode collector;
a coating portion having an active material coated on the electrode collector;
a first non-coating portion on which the active material is not applied to the electrode collector, the first non-coating portion being adjacent to the coating portion in a width direction perpendicular to a longitudinal direction of the electrode collector, the first non-coating portion having a longitudinal dimension extending along the longitudinal direction of the electrode collector, the first non-coating portion configured to be bent and electrically coupled to the can member; and
a second non-coating portion on which the active material is not applied to the electrode collector, the second non-coating portion being adjacent to the coating portion in the longitudinal direction.
2. The secondary battery of claim 1, wherein the second non-coating portion is located at a longitudinal end of the electrode collector.
3. The secondary battery of claim 1, wherein the first non-coating portion is spaced apart from the second non-coating portion.
4. The secondary battery of claim 3, wherein the first non-coating portion is spaced apart from the second non-coating portion along the longitudinal direction of the electrode collector.
5. The secondary battery of claim 1, wherein the electrode collector defines a first width in the width direction along the first non-coating portion in the longitudinal

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direction, and wherein the electrode collector defines a second width in the width direction along a segment of the electrode collector in the longitudinal direction that excludes the first non-coating portion, the first width being greater than the second width.

6. The secondary battery of claim 5, wherein the segment of the electrode collector is a remainder of the electrode collector in the longitudinal direction that excludes the first non-coating portion.

7. The secondary battery of claim 5, wherein the segment of the electrode collector extends along the second non-coating portion in the longitudinal direction.

8. The secondary battery of claim 1, wherein the first non-coating portion continuously extends across 1/3 or more of a total length of the coating portion in the longitudinal direction.

9. The secondary battery of claim 1, wherein the first non-coating portion continuously extends across at least two circumferential windings of the electrode collector in the longitudinal direction.

10. The secondary battery of claim 1, wherein the first non-coating portion is attached to the can member.

11. The secondary battery of claim 1, further comprising an electrode tab attached to and extending from the second non-coating portion.

12. The secondary battery of claim 1, wherein the first non-coating portion is configured such that an electric current in the first non-coating portion flows toward the can member in the width direction.

13. The secondary battery of claim 1, wherein the first non-coating portion is formed from a metal foil of the electrode collector.

14. The secondary battery of claim 1, wherein the can member has a cylindrical shape.

ABSTRACT

An electrode for a secondary battery may include an electrode collector, a coating portion having an active material coated on the electrode collector, a first non-coating portion on which the active material is not applied to the electrode collector, and a second non-coating portion on which the active material is not applied to the electrode collector. The first non-coating portion may be adjacent to the coating portion in a width direction perpendicular to a longitudinal direction of the electrode collector, the first non-coating portion having a longitudinal dimension extending along the longitudinal direction, the first non-coating portion configured to be bent and electrically coupled to a can member of the secondary battery. The second non-coating portion may be adjacent to the coating portion in the longitudinal direction. The electrode may be stacked into a stack with a separator and a second electrode of an opposite polarity than the electrode.

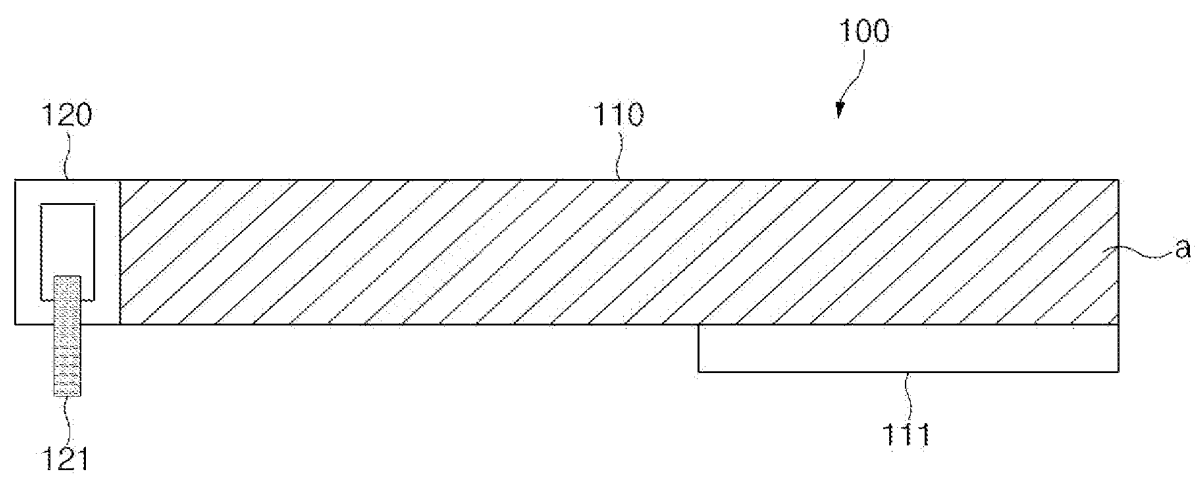


FIG. 1

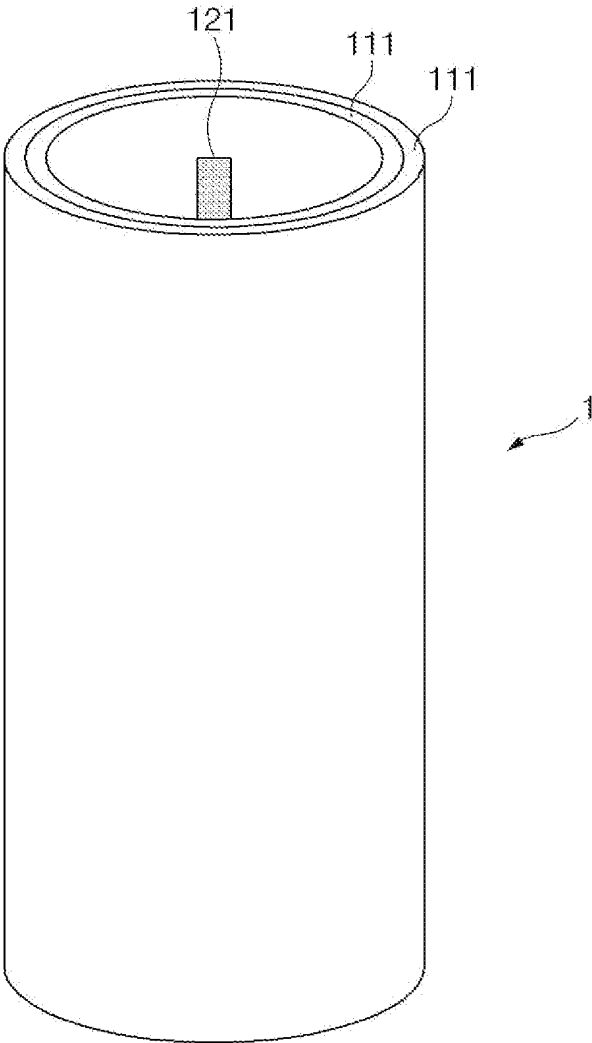


FIG.2

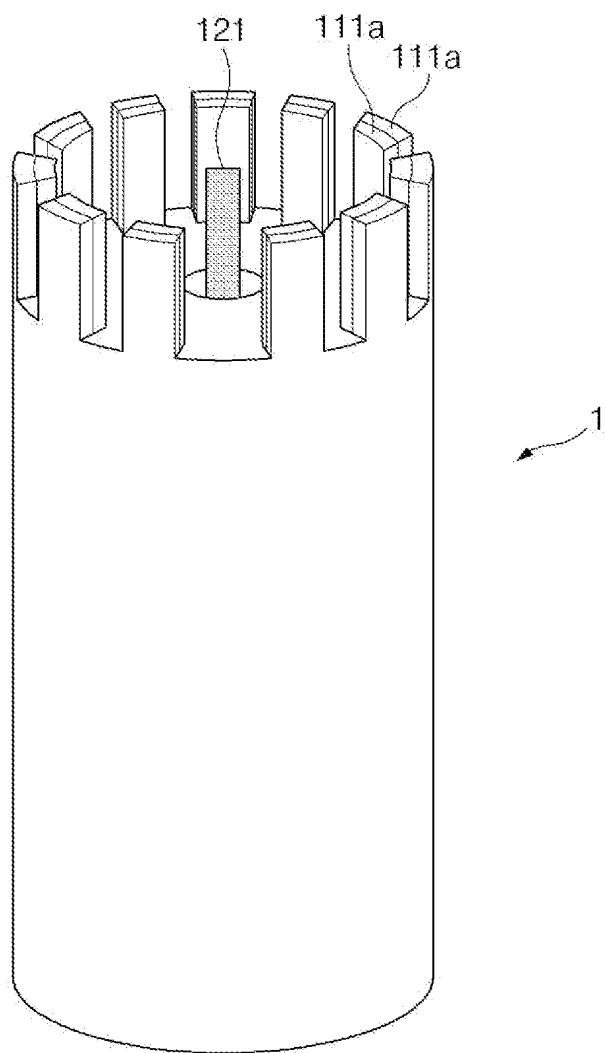


FIG. 3

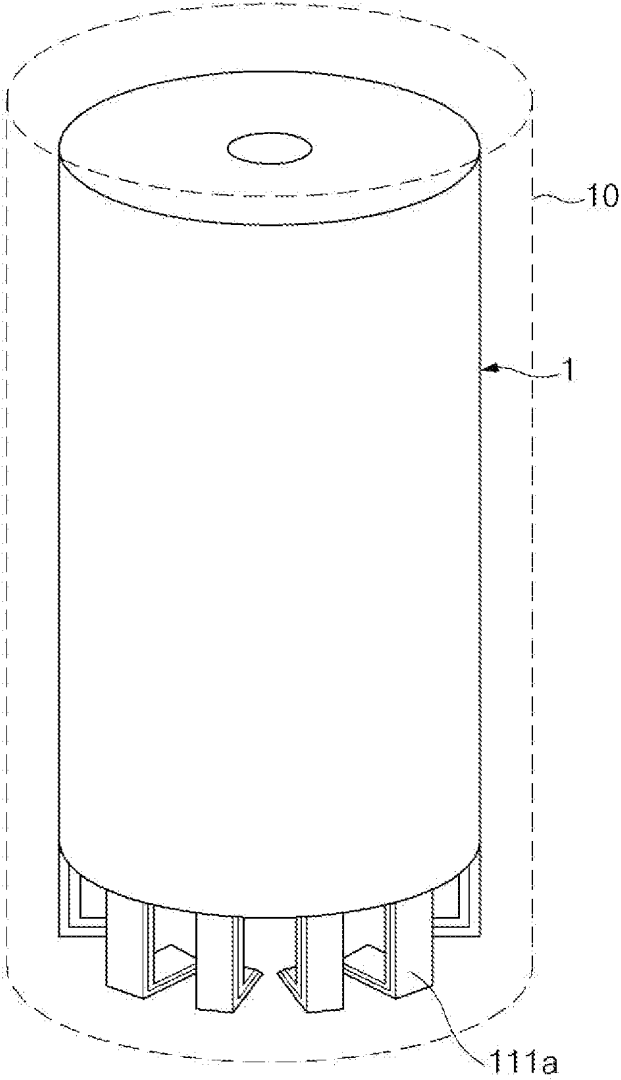


FIG. 4

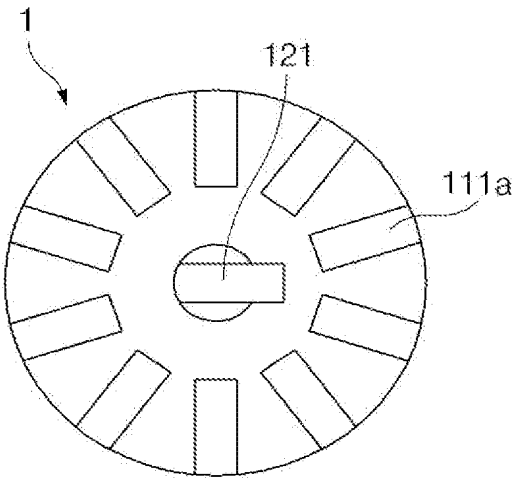


FIG. 5

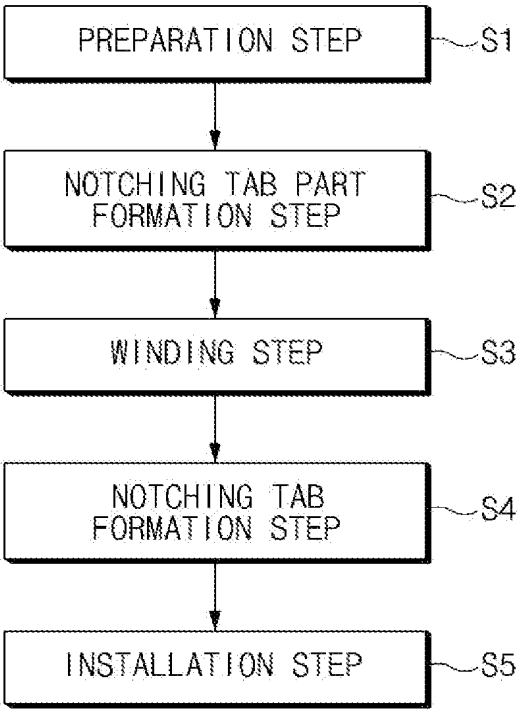


FIG.6

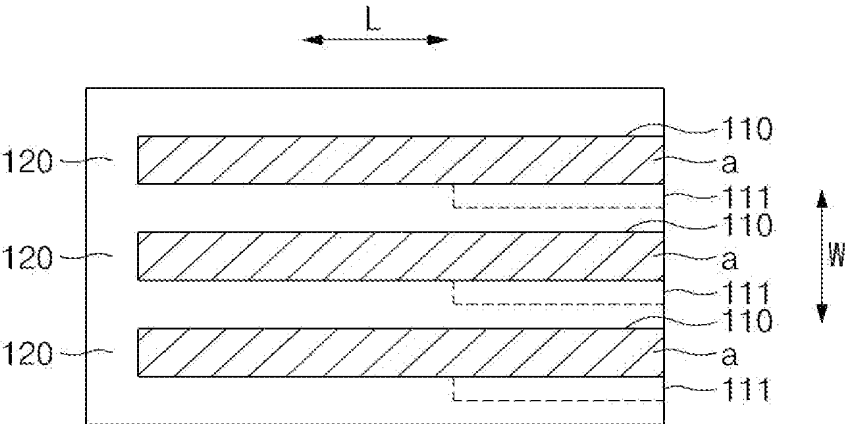


FIG. 7

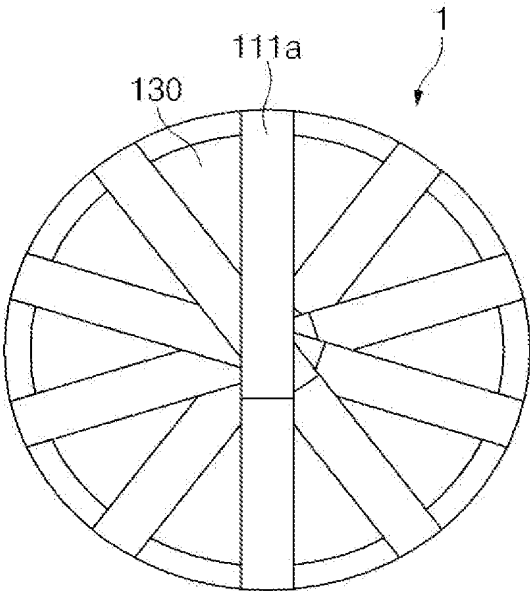


FIG. 8

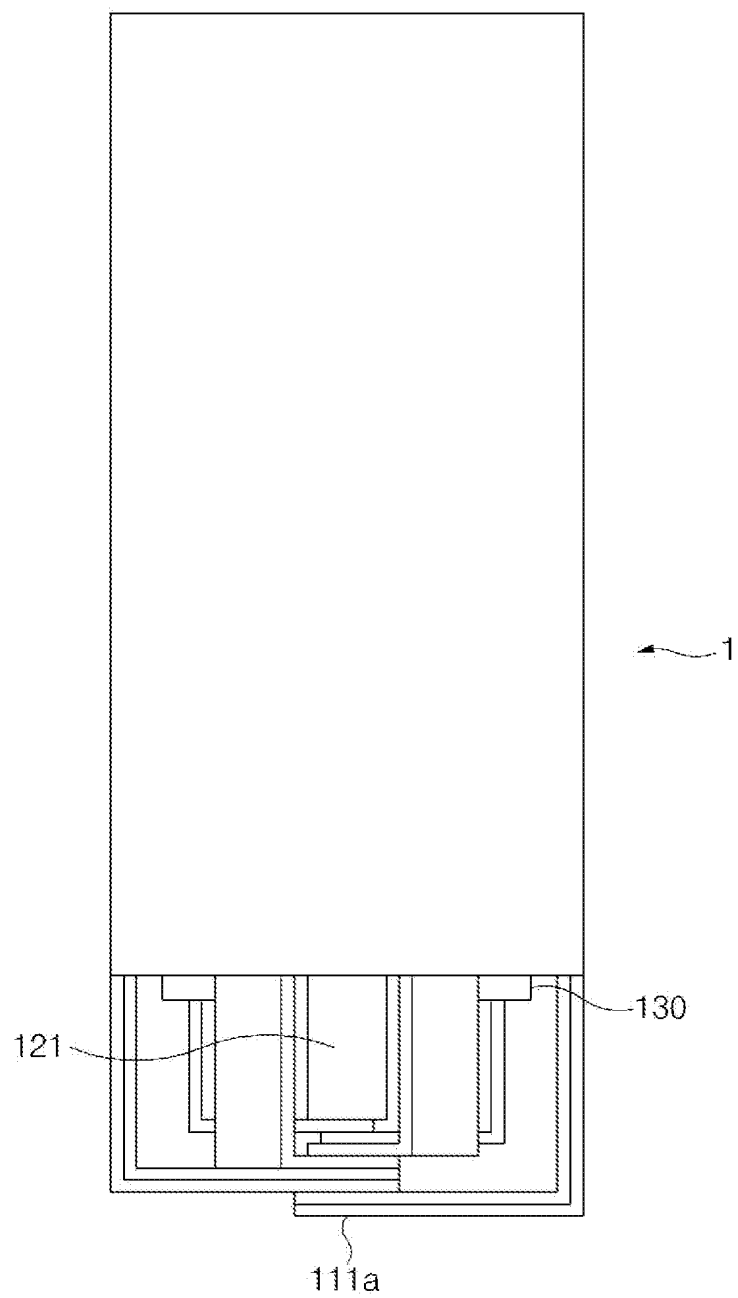


FIG. 9

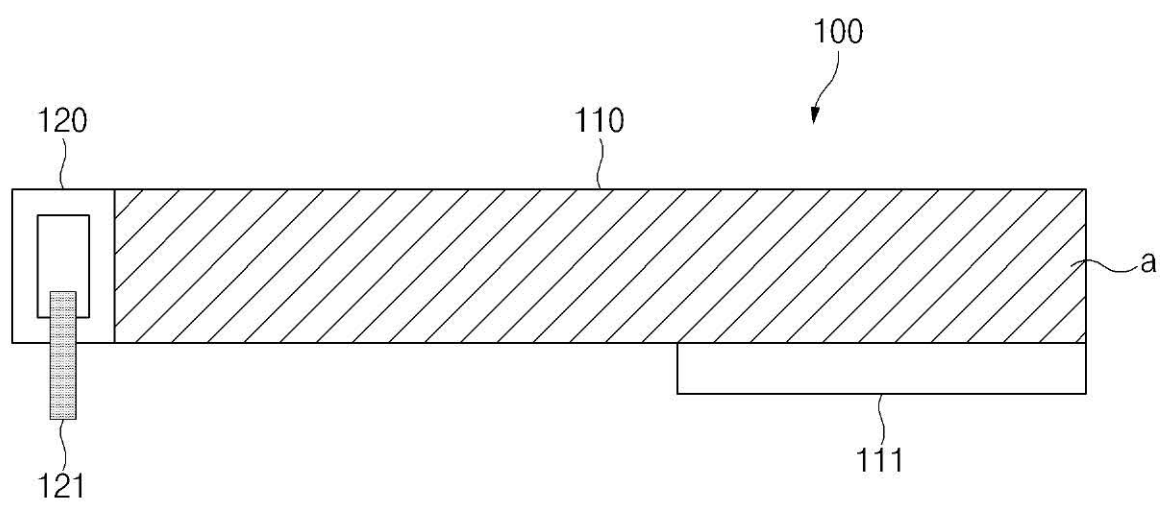


FIG.1

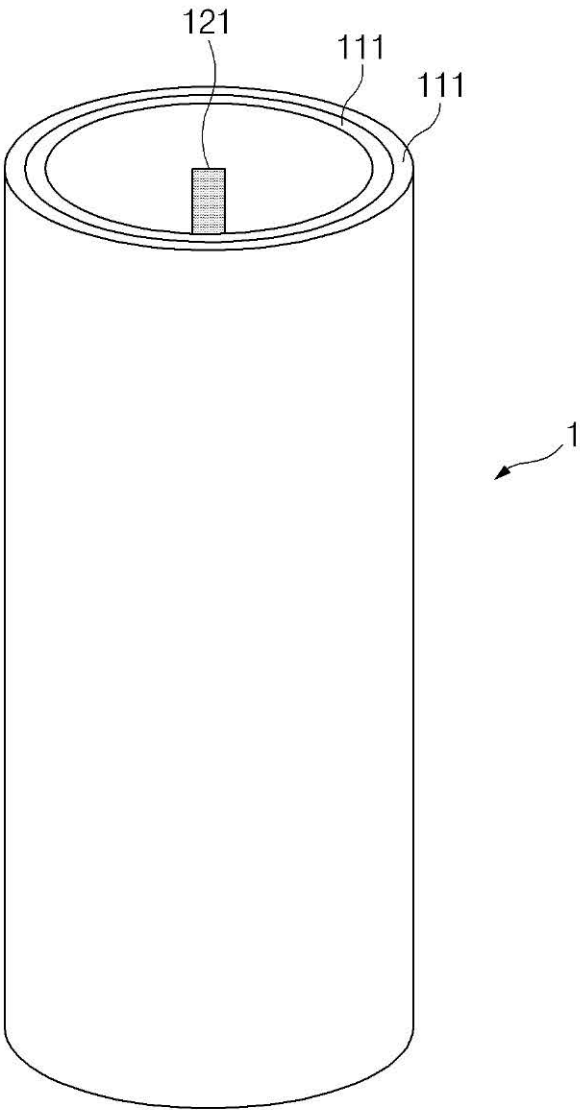


FIG.2

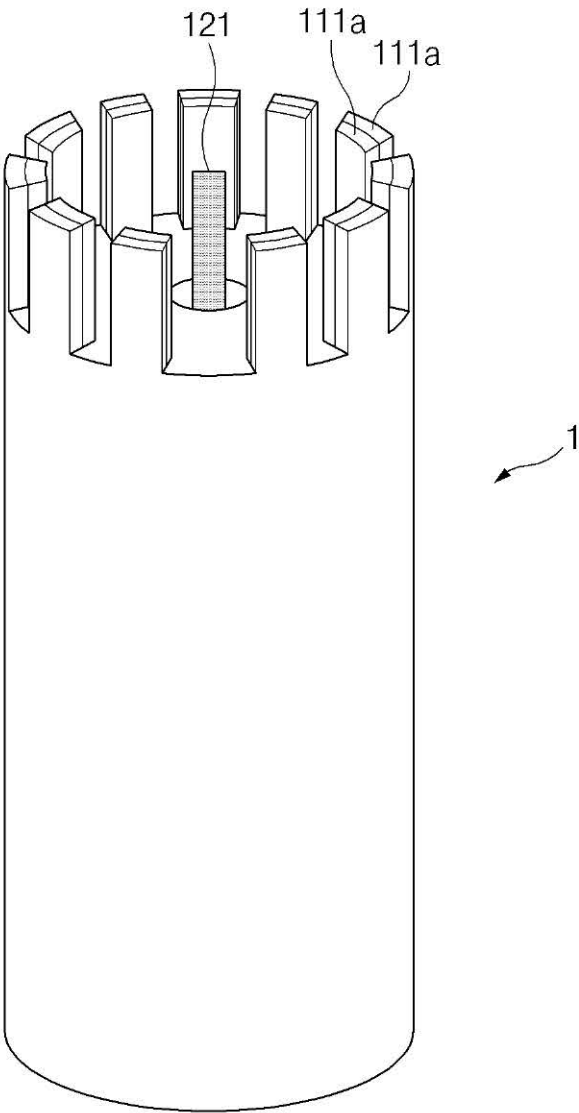


FIG.3

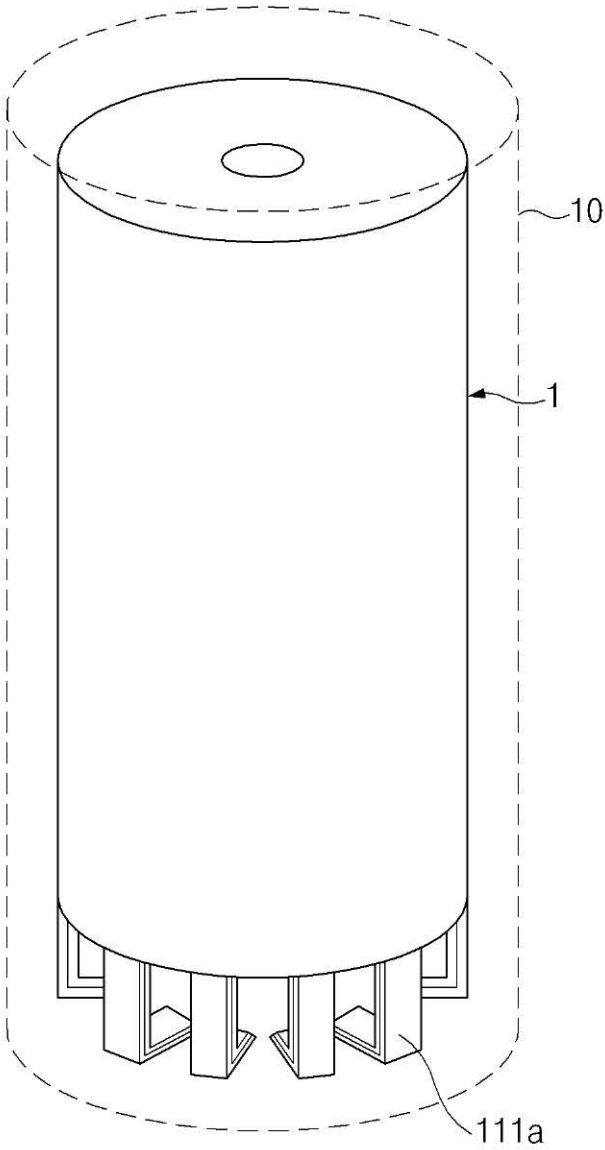


FIG.4

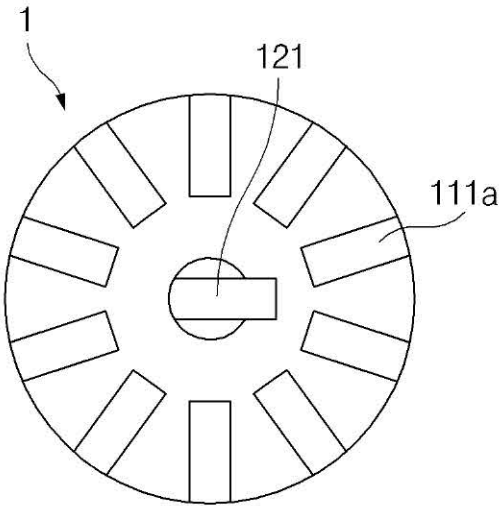


FIG. 5

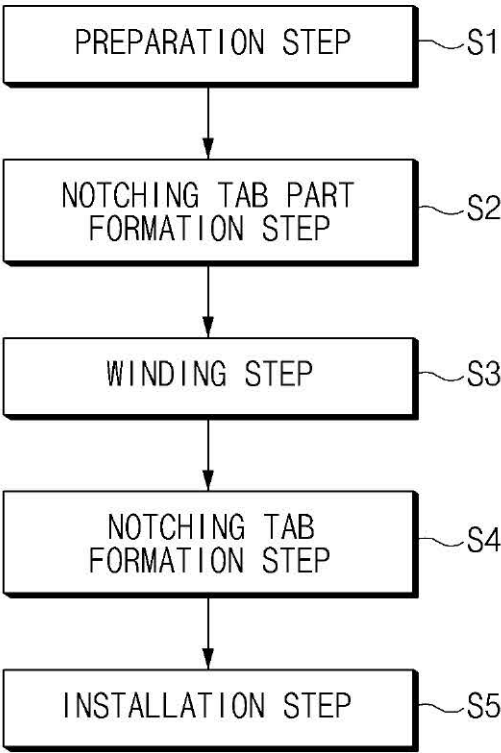


FIG.6

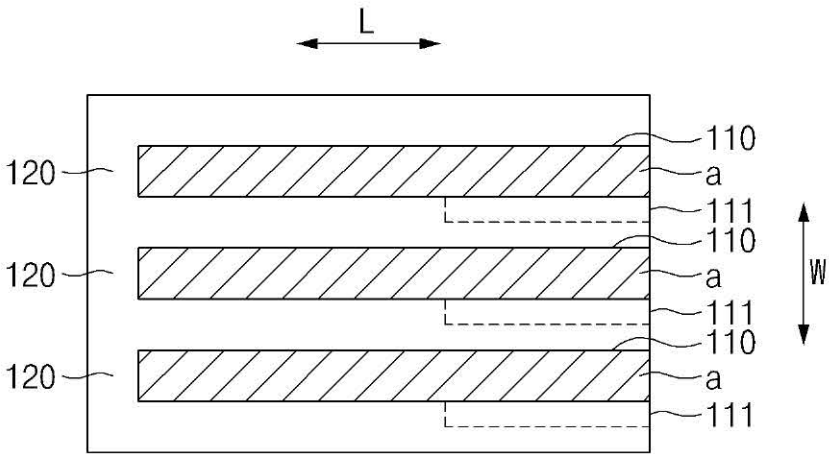


FIG.7

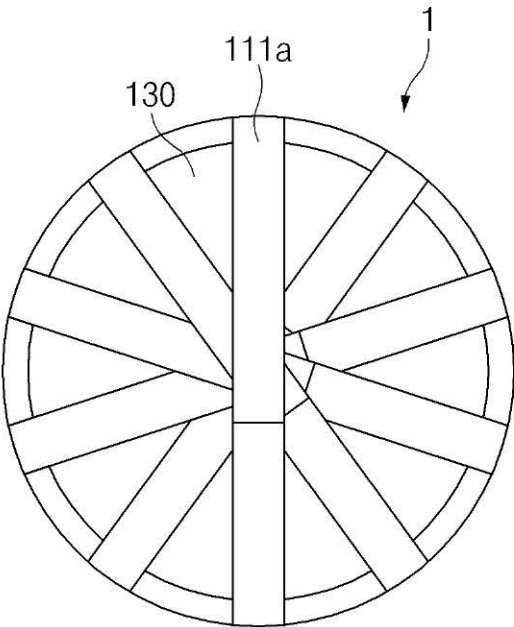


FIG. 8

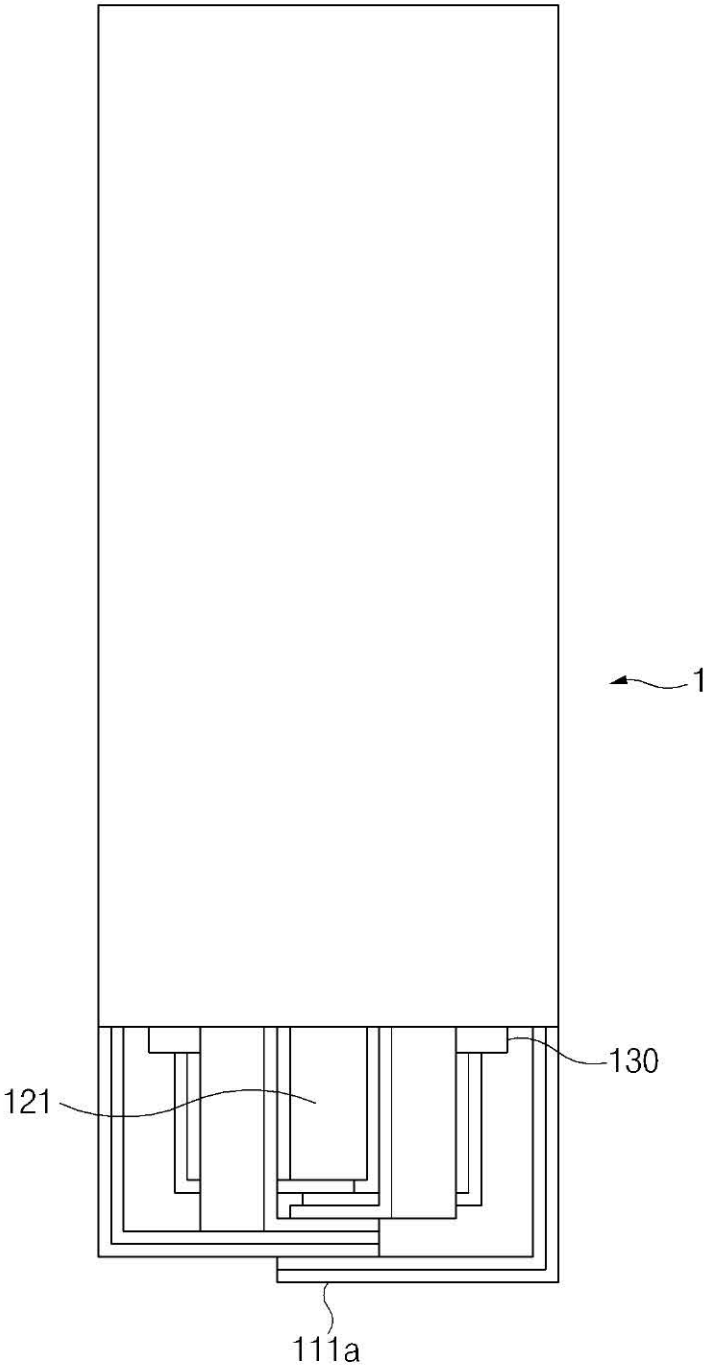


FIG.9



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APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/207,758	05/14/2025 01:37:38 PM Z ET	LGES 3.3F-1654 CCCC II

Title of Invention

Secondary Battery And Method For Manufacturing The Same

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	2553	FILED BY	Sharon McGee
PATENT CENTER #	70427322	AUTHORIZED BY	Kevin Kocun
CUSTOMER #	86765	FILING DATE	-
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Jae Woong Kim

Payment Information

PAYMENT METHOD	PAYMENT TRANSACTION ID	PAYMENT AUTHORIZED BY
CARD / 1060	E20255DD38168307	Sharon McGee

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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY.DOCKET.NO	TOT CLAIMS	IND CLAIMS
19/207,758	05/14/2025		2000	LGES 3.3F-1654 CCCC II	14	1

CONFIRMATION NO. 2553

86765
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Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

FILING RECEIPT



0000000089072527

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Assignment For Published Patent Application

LG Energy Solution, Ltd., Seoul, KOREA, REPUBLIC OF

Power of Attorney: The patent practitioners associated with Customer Number 86765

Domestic Applications for which benefit is claimed

This application is a CON of 18/620,269 03/28/2024
which is a CON of 18/618,609 03/27/2024
which is a CON of 18/132,719 04/10/2023 PAT 11,973,177
which is a CON of 16/498,751 09/27/2019 PAT 11,652,232
which is a 371 of PCT/KR2018/004366 04/13/2018

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)
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Projected Publication Date: 08/28/2025

Non-Publication Request: No

Early Publication Request: No

Title

Secondary Battery And Method For Manufacturing The Same

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
19/207,758	05/14/2025	Jae Woong Kim	LGES 3.3F-1654 CCCC II

CONFIRMATION NO. 2553

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Cranford, NJ 07016

PUBLICATION NOTICE



Date Mailed: 08/29/2025

Title:Secondary Battery And Method For Manufacturing The Same

Publication No.US-2025-0273726-A1

Publication Date:08/28/2025

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The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

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APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/207,758	01/16/2026 11:23:23 AM Z ET	LGES 3.3F-1654 CCCC II

Title of Invention

Secondary Battery And Method For Manufacturing The Same

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	2553	FILED BY	Sharon McGee
PATENT CENTER #	74015341	FILING DATE	05/14/2025
CUSTOMER #	86765	FIRST NAMED INVENTOR	Jae Woong Kim
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Kevin Kocun

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
LGES6209 - Information Disclosure Statement Fillable PDF_1.pdf	5	Information Disclosure Statement (IDS) Form (SB08)	1314 KB
IDS - Information Disclosure Statement - Reference Enclosed LD-458_1.pdf	1	Transmittal Letter	23 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
LGES6209 - Information Disclosure Statement Fillable PDF_1.pdf	EE0E89D3DAE47C2E1092FC4CB1E43585237CC4C09322A5C93988FFBB96B45BC3600D8FBB250695977F6AC6CD2955AA9F8CA6932CEB54B6DFF7F3F14C26CBD185

IDS - Information Disclosure	2E6968E08568786B66C7531E5571E80EB13F72821E8AC5BB1
Statement - Reference	A9381BE5DD29148011656A262BCD001A31CC1433634D72130
Enclosed LD-458_1.pdf	9D91207BAEE278909ACC0BCA3454E3

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT

Application Number		19207758
Filing Date		2025-05-14
First Named Inventor	Jae Woong Kim	
Art Unit		N/A
Examiner Name	Not Yet Assigned	
Attorney Docket Number		LGES 3.3F-1654 CON CON CO

TIMING STATEMENT

Please see 37 CFR 1.97(e) to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

- ☐ See attached timing statement under 37 CFR 1.97(e).
- ☒ A timing statement under 37 CFR 1.97(e) is not submitted herewith.

TIMING FEE

☐ The fee set forth in 37 CFR 1.17(p) has been submitted herewith.

COPIES

☐ An identification of an earlier application pursuant to 37 CFR 1.98(d)(1) is attached.

SIZE FEE ASSERTION

Please see 37 CFR 1.17(v) and the [IDS Auto-Load Instructions](#) for completing this form to make the appropriate selection of an assertion under 37 CFR 1.98. For the information disclosure statement (IDS) submitted herewith, the applicant or patent owner certifies the following with respect to the cumulative number of applicant-provided or patent-owner provided items of information submitted to date including those in the accompanying IDS (select only one):

- ☒ No IDS size fee is required under 37 CFR 1.17(v) at this time.
- ☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(1).
- ☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(2).
- ☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(3).

SIGNATURE

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 11.18. Please see 37 CFR 1.4(d) for the form of the signature.

Signature	/Joseph H. Karlin/	Date (YYYY-MM-DD)	2026-01-16
Name/Print	Joseph H. Karlin	Registration Number	62,471

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT

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This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		19207758	
	Filing Date		2025-05-14	
	First Named Inventor	Jae Woong Kim		
	Art Unit		N/A	
	Examiner Name	Not Yet Assigned		
	Attorney Docket Number		LGES 3.3F-1654 CON CON CO	

U.S. PATENTS							Remove	
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Inventor of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear		
	1	12334493		2025-06-17	Kim et al.			
If you wish to add additional U.S. Patent citation information, please click the Add button.							Add	
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	Attorney Docket Number		LGES 3.3F-1654 CON CON CO

	1		☐
If you wish to add additional non-patent literature document citation information, please click the Add button Add			
EXAMINER SIGNATURE			
Examiner Signature		Date Considered	
*EXAMINER: Initial if reference considered and whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.			
¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.			

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: January 16, 2026

Electronic Signature for Joseph H. Karlin: /Joseph H. Karlin/

Docket No. LGES 3.3F-1654 CON CON CON CON II
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Kim et al.

Application No.: 19/207,758

Confirmation No.: 2553

Filed: May 14, 2025

Art Unit: N/A

For: Secondary Battery And Method For
Manufacturing The Same

Examiner: Not Yet Assigned

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Dear Commissioner:

It is respectfully requested that the reference(s) listed on the enclosed form be made of record and considered with respect to the above-referenced U.S. patent application. A copy of each reference which is not a U.S. patent or patent application is enclosed. Submission of the present Information Disclosure Statement should not be taken as an admission that the cited reference(s) is/are legally available prior art or that the same is/are pertinent or material or even analogous art to the claimed invention.

Applicant believes no fee is due with this response. However, if a fee is due, please charge our Deposit Account No. 12-1095, under Order No. LGES 3.3F-1654 CON CON CON CON II from which the undersigned is authorized to draw.

Dated: January 16, 2026

Respectfully submitted,

Electronic signature: /Joseph H. Karlin/
Joseph H. Karlin
Registration No.: 62,471
LERNER DAVID LLP
20 Commerce Dr.
Cranford, New Jersey 07016
(908) 654-5000
Attorney for Applicant(s)

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875						Application or Docket Number 19/207,758													
APPLICATION AS FILED - PART I																			
(Column 1)			(Column 2)			SMALL ENTITY		OR		OTHER THAN SMALL ENTITY									
FOR		NUMBER FILED		NUMBER EXTRA		RATE(\$)		FEE(\$)		RATE(\$)		FEE(\$)							
BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A				N/A		350							
SEARCH FEE (37 CFR 1.16(k), (l), or (m))		N/A		N/A		N/A				N/A		770							
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A				N/A		880							
TOTAL CLAIMS (37 CFR 1.16(i))		14 minus 20 =		* 0						x =		0							
INDEPENDENT CLAIMS (37 CFR 1.16(h))		1 minus 3 =		*						x =		0							
APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).										0							
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))												0							
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL				TOTAL		2000							
APPLICATION AS AMENDED - PART II																			
(Column 1)			(Column 2)			(Column 3)			SMALL ENTITY		OR		OTHER THAN SMALL ENTITY						
AMENDMENT A				CLAIMS REMAINING AFTER AMENDMENT				HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE(\$)		ADDITIONAL FEE(\$)		RATE(\$)		ADDITIONAL FEE(\$)	
		Total (37 CFR 1.16(i))		*		Minus		**		=		x =				OR		x =	
		Independent (37 CFR 1.16(h))		*		Minus		***		=		x =				OR		x =	
		Application Size Fee (37 CFR 1.16(s))												OR					
		FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))												OR					
TOTAL ADD'L FEE												OR		TOTAL ADD'L FEE					
(Column 1)			(Column 2)			(Column 3)			SMALL ENTITY		OR		OTHER THAN SMALL ENTITY						
AMENDMENT B				CLAIMS REMAINING AFTER AMENDMENT				HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE(\$)		ADDITIONAL FEE(\$)		RATE(\$)		ADDITIONAL FEE(\$)	
		Total (37 CFR 1.16(i))		*		Minus		**		=		x =				OR		x =	
		Independent (37 CFR 1.16(h))		*		Minus		***		=		x =				OR		x =	
		Application Size Fee (37 CFR 1.16(s))												OR					
		FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))												OR					
TOTAL ADD'L FEE												OR		TOTAL ADD'L FEE					
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.																			